



OFFICIAL

Gestational Diabetes Mellitus and Gestational Diabetes Mellitus Requiring Pharmacological Therapy Guideline

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1. Purpose

The purpose of the Gestational Diabetes Mellitus and Gestational Diabetes Mellitus Requiring Pharmacological Therapy Guideline is to provide a standardised, evidence-based framework to support the safe, effective, and consistent care of women diagnosed with GDM across Armadale Kalamunda Group (AKG). This guideline aims to enhance multidisciplinary collaboration, streamline clinical workflows, and ensure alignment with current best practice. It serves as a practical reference for healthcare staff to guide decision-making, promote optimal maternal and neonatal outcomes, and facilitate the delivery of patient-centred care throughout the antenatal, intrapartum, and postnatal periods.

AKG is committed to providing women and their neonates with coordinated, evidence-based, and culturally respectful care, with consideration of the women's rights, dignity, and consent.

Note: In this guideline the term women refers to all people who are pregnant or birthing including those who identify as transgender, non-binary, or gender diverse.

2. Background

Gestational Diabetes Mellitus (GDM) is one of the most prevalent complications of pregnancy in Australia. During the 2020–2022 period, just over 19% of all pregnancies were affected by GDM.¹ This condition significantly increases a woman's lifetime risk of developing type 2 diabetes—by approximately 22-fold—and is associated with an elevated risk of adverse perinatal outcomes.^{1 2}

3. Applicability

This guideline applies to all Medical, Nursing, Midwifery and Allied Health staff within AKG providing care for patients with GDM requiring pharmacological treatment.

4. Guidelines

This guideline aims to:

- Promote early screening and timely diagnosis of GDM and Overt Diabetes and women requiring pharmacological treatment.
- Promote evidence-based, interprofessional maternity care
- Assist in the detection, monitoring and management of complications associated with GDMi, ensuring timely intervention, management and escalation when required.
- Provide consistent and appropriate care for women and their neonates at risk of maternal and fetal/neonatal complications due to GDM.
- Offer a structured, culturally safe education pathway covering diet, physical activity, medication, self-care, and blood glucose level (BGL) monitoring.
- Improve maternal and neonatal outcomes through women centred, targeted interventions and coordinated multidisciplinary support
- Optimise glycaemic control throughout pregnancy.
- Prevent or reduce the need for hospitalisation of women and their neonates through proactive stabilisation and ongoing support.
- Identify women with persistent diabetes or carbohydrate intolerance post-pregnancy and ensure appropriate follow-up and long-term management.

5. Definition

GDM is defined as glucose intolerance first recognised during pregnancy that does not fulfil criteria for overt diabetes in pregnancy (DIP). ¹ It is distinct from pre-existing diabetes and requires separate diagnostic and management pathways.

Early gestational diabetes mellitus refers to GDM detected before 20 weeks' gestation.

Table 1 Types of Diabetes and Definitions

Types of diabetes	Definition
Gestational Diabetes Mellitus (GDM)	Glucose intolerance with onset or first recognition during pregnancy that does not meet criteria for overt diabetes in pregnancy. When a woman has GDM in a pregnancy there is a higher risk of it occurring in a subsequent pregnancy and of developing type 2 diabetes mellitus in later life.
Pre-existing Type 1 Diabetes	Diabetes diagnosed prior to pregnancy, characterised by absolute insulin deficiency due to autoimmune destruction of pancreatic beta cells.
Pre-existing Type 2 Diabetes	Diabetes diagnosed prior to pregnancy, because of pancreas not being able to produce sufficient insulin to overcome the insulin resistance that occurs in this condition.
Overt Diabetes in Pregnancy	Diabetes first detected during pregnancy that meets standard (non-pregnant) diagnostic criteria for diabetes: fasting plasma glucose ≥ 7.0 mmol/L, random plasma glucose ≥ 11.1 mmol/L, or HbA1c $\geq 6.5\%$ (48 mmol/mol), typically identified early in pregnancy. A definitive diagnosis of non-gestational diabetes cannot be made until after pregnancy These women may have higher risk of their glucose intolerance not normalising after pregnancy.
Maturity Onset Diabetes of the Young (MODY)	A monogenic form of diabetes, usually diagnosed before pregnancy, sometimes discovered during pregnancy; caused by a gene mutation

6. Management of Pregnant Women with Diabetes in AKG ED or AKG ICU

If a woman who is pregnant and diagnosed with diabetes presents to the Emergency Department (ED) or is admitted to the Intensive Care Unit (ICU), after completing standard admission processes, including consulting with Specialist Obstetrician and GPO per BAU, the following steps must be followed:

- Complete 4-point blood glucose (BGL) profiles daily.
- Document all readings on AKMR60.8.3 Obstetric Subcutaneous Insulin Order and in the Digital Medical Record (DMR).
- Target BGL Range:
 - Follow the target blood glucose levels in pregnancy as outlined on AKMR60.8.3 and in this guideline.
 - Type 1 women require immediate consultation with Obstetric physician at Women's and Newborn Services (KEMH) for guidance on BGL range.
- Escalation of Care:
 - If BGLs fall outside the target range, contact the AHS Diabetes Service or KEMH Obstetric Physician for further management advice.

Contact the Obstetric Physician at KEMH by calling the switchboard on 6458 2222 and ask for Obstetric Physician.

Any pregnant woman with DKA in ED/ICU → Immediately contact the WNHS/KEMH Obstetric Physician (24/7) for care plan development and transfer once stabilised.

Refer to [Appendix 1](#) Presentation and admission for non-obstetric condition of women living with Diabetes who is pregnant flowchart.

NB: WNHS/KEMH are the only health services that provides care for women living with Type 1 & 2 diabetes who are pregnant. FSH and WNHS are the tertiary care providers for overt diabetes and for those women who are living with GDM on treatment who fall outside of AKG's scope of practice.

7. Screening Recommendations

All pregnant women without known pre-existing diabetes should undergo universal screening for GDM. Early testing in the first trimester is crucial to identify GDM promptly, enabling timely intervention and reducing the risk of adverse maternal and fetal outcomes. Early screening is also used to identify overt diabetes - these women will be transferred to either FSH or KEMH for tertiary level care.

7.1 High-Risk Women

Women with one or more risk factors should be offered an early HbA1c test - completed within the first trimester.

7.2 Pre-Existing Diabetes

Women with known diabetes require referral to WHNS for comprehensive antenatal, intrapartum, and postnatal care.

7.3 Diagnosis

Diabetes in pregnancy includes both:

- Women with previously undiagnosed abnormalities of glucose tolerance.
- Women with glucose abnormalities that arise during pregnancy alone.

[Appendix 2](#) provides guidance for clinicians on the screening process.

7.4 Exclusion Criteria

AKG GDM Midwifery and Postnatal service (MAPS) does not provide management for the following:

- Women with pre-existing (pregestational) diabetes.
- Women with an overt diabetes diagnosis.
- Women with a BMI above 38 at booking or BMI ≥ 40 during pregnancy.

7.5 Screening Gestation

- Low-risk women: Offer screening at 24–28 weeks gestation.
- Increased risk women: Require early screening (see section 7.7).

7.6 Screening test

GDM is diagnosed if one or more of the following are met:

- Fasting glucose ≥ 5.3 – 6.9 mmol/L
- 1-hour glucose ≥ 10.6 mmol/L
- 2-hour glucose ≥ 9.0 – 11.0 mmol/L

Alternatives:

- If OGTT not tolerated: GDM diagnosed if fasting plasma glucose ≥ 5.3 mmol/L.
- Non-fasting random plasma glucose: If ≥ 7.8 but < 11.1 mmol/L, proceed to OGTT. ≥ 11.1 mmol/L is diagnostic.

7.7 Screening Recommendation - High risk of GDM

- Offer HbA1c in the first trimester
- HbA1c $\geq 6.4\%$ should be offered 75g GTT
- If negative, repeat 75 g OGTT at 24–28 weeks

7.8 Risk Factors for GDM

Women are at greater risk of developing GDM if they have the following risk factors:

- Previous GDM
- Previously elevated blood glucose level at any time, past impaired glucose tolerance
- Ethnicity: Asian, Indian subcontinent, Aboriginal, Torres Strait Islander, Pacific Islander, Māori, Middle Eastern, non-white African
- Maternal age ≥ 40 years
- Family history of DM (1st degree relative with diabetes mellitus including a sister with GDM)
- Pre-pregnancy BMI > 30 kg/m²
- Hypertension prior to 20 weeks
- Previous macrosomia (baby with birth weight more than 4000g)
- Previous adverse pregnancy outcome suggestive of undiagnosed GDM e.g. shoulder dystocia, unexplained stillbirth/perinatal loss
- Multiple pregnancy- Twins or higher

- Previous baby with congenital abnormalities
- Polycystic ovarian syndrome
- Medications: corticosteroids, antipsychotics

7.9 Woman Declining Screening

If a woman declines screening and has a previous history of GDM or multiple risk factors, consideration needs to be given to offering BGL monitoring and diagnosis without screening. If a GDM diagnosis is suspected, or if completed HbA1c $\geq 6.0\%$, or clinical observation supports possible GDM diagnosis consider a trial of self-BGL testing.

If a woman vomits after drinking glucose, use fasting PGL or if the women decline the OGTT offer fasting PGL, if in the first trimester offer HbA1c.

HbA1c is only suitable in the first trimester before erythrocytes formation starts to increase, and basal glucose levels fall.

Note: HbA1c may be inaccurate in conditions that affect red blood cells, such as anaemia, haemoglobinopathy, kidney or liver disease or recent iron/ blood transfusions. Caution should be exercised with such known factors.³

7.10 Repeat screening and/or late testing for GDM in Pregnancy

A repeat oral glucose tolerance test (OGTT) in the third trimester (after 28 weeks) should only be considered when there is a strong clinical suspicion of GDM, such as large-for-gestational-age fetus (LGA), accelerated fetal growth, polyhydramnios, or persistent glucosuria.

- HbA1c is only recommended for GDM in the first trimester of pregnancy due to its low sensitivity³. This limitation is primarily attributed to the physiological decline in HbA1c levels during the second trimester, which may result in an underestimation of glycaemic status³.
- OGTTs performed after 33+6 weeks gestation are not recommended, as results are unlikely to influence pregnancy management.

Where GDM is suspected and OGTT is no longer appropriate, diagnosis may be made based on the clinical picture. In such cases, the notification form on the NDSS Health Professional Portal must be completed by the appropriate healthcare professional with the patients consent and documented in the DMR. An immediate e-referral to the Armadale Health Service (AHS) Diabetes Service is required.

7.11 Maternal Medications and Screening

Steroid Use and OGTT:

- OGTT should not be performed within one week of steroid use.

Metformin Use:

- Metformin use for Polycystic Ovarian Syndrome (PCOS) may affect oral glucose tolerance test (OGTT) results and should be ceased one week prior to GDM screening, see: [AKG Antenatal Clinical Guideline](#).
- Metformin is not recommended during pregnancy unless required for glycemic control when insulin is not tolerated. Women using metformin for PCOS should discontinue use in the first trimester.

7.12 Screening for GDM after bariatric surgery

Post-bariatric surgery patients may experience dumping syndrome after consuming high-glycaemic carbohydrates, making the standard 75-g oral glucose tolerance test (OGTT) poorly

tolerated. Symptoms include gastro-intestinal upset, late hypoglycaemia, tachycardia, palpitations and sweating.

Recommended Approach:

- Avoid OGTT if poorly tolerated; consider ordering fasting plasma glucose and HbA1c as alternatives.
- Screen for diabetes in early pregnancy (first or early second trimester) and again in the third trimester due to changes in glucose metabolism.

Clinical Considerations:

- OGTT tolerance depends on type and timing of surgery.
- Use early fasting plasma glucose to detect pre-existing or early GDM.
- Monitor and document fetal growth as outlined in the [AKG Antenatal Clinical Guideline](#).

Key Points for Clinicians:

- Assess individual tolerance for OGTT. (See Table 3 Bariatric surgery and tolerance to OGTT)
- Consider alternative GDM screening methods.
- Repeat screening later in pregnancy due to hormonal changes

Table 2 Bariatric surgery and tolerance to OGTT

Type of Bariatric Surgery	Tolerance of the OGTT	
Lap Band/Orion Balloon	Most women tolerate the OGTT well	
Gastric Sleeve	Less than 12-18 months since surgery	OGTT may be poorly tolerated
	More than 12-18 months since surgery	OGTT normally well tolerated
Bypass surgery	Most women can NOT tolerate the OGTT	

Table 3 Diagnosis criteria for women who had Bariatric surgery and unable to tolerate OGTT

Gestation	Test	Diagnosis
Booking visit (all women)	Fasting Plasma PGL	≥ 5.3 – Diagnostic of GDM
24-28/40 if previous FPGL was normal	Repeat Fasting Plasma PGL	≥ 5.3 – Diagnostic of GDM
If at any gestation after 28/40 baby is showing signs of being LGA, women will need an extra growth scan.	Random finger prick BGLs at clinics. Contact Dietitian or Diabetes Educator or Diabetes Educator Midwife if RBGLs are >6.7.	

For more information or advice, contact Diabetes Service on 9391 1111

8. Diagnosis Criteria for GDM

A diagnosis of GDM is confirmed if one or more of the following plasma glucose values are elevated:

Table 4 Diagnostic Criteria for GDM

The 2025 Australasian Diabetes in Pregnancy Society (ADIPS) guidelines for the diagnosis of Gestational Diabetes mellitus (GDM) using the pregnancy oral glucose tolerance test (POGTT) are a venous plasma glucose (PG) level of:

Fasting venous PG	≥5.3 mmol/L
1-Hour venous PG	≥10.6 mmol/L
2-hour venous PG	≥9.0 mmol/L

- HbA1c ≥ 6.0-6.5% +/- fasting ≥ 5.3 in early pregnancy

The diagnosis of GDM is made if one or more of these that are outside of this criterion²

Note: Under the new ADIPS criteria, some women with a history of GDM may now be considered at low risk of developing GDM in a subsequent pregnancy. However, if a woman expresses concerns about possible adverse outcomes, clinicians must provide support and offer individualised care tailored to her needs during the current pregnancy^{2,3}.

Once pharmacological treatment has commenced titration of insulin will be from fasting 5.1mmol/L.

8.1 Likely overt diabetes

Women with markedly elevated glucose levels in pregnancy (fasting >7.0 mmol/L, random BSL >11.1 mmol/L, or HbA1c >6.5%) are considered to have likely overt diabetes as formal diagnosis cannot occur until the postpartum period. Overt diabetes during pregnancy can increase the risk of complications and therefore all women who are considered to have overt diabetes must have their pregnancy care transferred to WHNS or FSH.

If a woman is suspected as having likely overt diabetes, urgently discuss with Myrtle Diabetes Educator / AKG Diabetes Service Clinical Nurse Specialist (CNS) for referral and transfer to appropriate tertiary site.

8.2 Criteria for Transfer of Women with GDM to Tertiary care.

Women with the following results will need their pregnancy, labour & birth and postnatal care transferred to either WHNS or FSH (based on patient post code). Referrals are to be made via e-Referral.

Criteria for referral to tertiary hospital:

- HbA1c ≥ 6.5% (48mmol/mol)
- Fasting BGL ≥ 7.0mmol/L
- OGTT 2-hour result ≥11.1mmol/L
- Other clinician concerns- to be discussed with an AKG diabetes endocrinologist or DS DE, Myrtle DE prior to referral.

Women with BMI > 38 on booking require referral to FSH/KEMH for birthing.

9. Complications of Undiagnosed GDM

Table 5 Complications of Undiagnosed GDM

Risk Type	Fetal	Maternal
Growth Issues	IUGR, Macrosomia	Obesity
Birth Complications	Shoulder dystocia, Birth trauma	Operative birth, Caesarean birth
Metabolic Issues	Polycythaemia, Hypoglycaemia	Hyperglycaemia
Respiratory & Other Conditions	Respiratory distress, Jaundice	Infection
Severe Outcomes	Fetal death (late, uncontrolled GDM)	Postpartum haemorrhage
Long-Term Risks	Glucose intolerance, Type 2 diabetes, Obesity	Recurrent GDM, Type 2 diabetes, Cardiovascular, Metabolic, Renal disorders

10. Antenatal Care Management

10.1 Referral to Diabetes Service

Referral to the AHS Diabetes Service should occur within 48 hours of receiving a positive POGTT result.

- An e-referral must be submitted by the healthcare practitioner reviewing the results to the diabetes service
- Document in the woman's handheld WA pregnancy record the pathology results and that e-referral has been sent.
- Document on the woman's DMR record:
 - Discussion with the woman
 - E-Referral Sent to Diabetes Services
 - Pathology Results

If the woman has not been informed of the results and is not present, the reviewing clinician should contact the woman to advise her of the result and document the discussion as outlined above. Consent should be gained for referral to NDSS.

Insulin requirements typically rise from 17–18 weeks and stabilise by approximately 36 weeks. A decrease in insulin requirements at or beyond 36 weeks requires urgent review by a Diabetes Educator, with consideration for referral to an Obstetric Consultant, Endocrinologist, or Obstetric Physician.

10.2 Initial education post diagnosis

The DE will review the initial GTT results and determine which type of education to offer the woman.

- Education will be provided via group or individual sessions within 10 days of referral.

- All women will have access to individualised education session taking into consideration cultural requirements, learning ability, family and social circumstances.
- Partners and family members should be encouraged and supported to attend.
- DE's working within the GDM MAPS service will provide education as deemed appropriate by the AHS Adult Diabetes Service (AHS ADS).
- Women will be reviewed in ANC/MAPS within 2 weeks of education for a Dietitian review.

10.3 Continuing education

- Continuing education sessions will be arranged by the Myrtle DE/Midwife, or the Dietitian as required.
- Classes are held regularly each week, with additional sessions available as required.
- Women receiving care through MAPS will receive ongoing education with the DE/midwife throughout their pregnancy.
- If further education is required, it can be arranged through the AHS ADS bookings are essential. Alternative formats for education delivery are also available; please contact the Diabetes Service for more information.

11. Clinical Responsibilities in the Management of GDM

Table 6 Clinician Responsibilities in the Management of GDM

Role	Key Responsibilities
All Healthcare Professionals	• Educate on the purpose, process, timing and implications of • the POGTT and document in the DMR. • Issue pathology request and document in DMR. • Refer all women with a positive POGTT to AHS Diabetes • Service within 48 hrs via e-Referral. • Ensure woman is informed of diagnosis and document in the DMR. • Refers to specialist clinic via e-Referral as needed and documents plan in the HHR and DMR. • Support timely triage by Diabetes Service. • Ensure the woman is aware her neonate requires BGL monitoring for the first 24 hours post-birth.
Endocrinologists / Obstetric Physicians	• Liaise with the broader care team inc. obstetricians, midwives and General Practitioner Obstetrics (GPO). • Prescribe and adjust pharmacological treatments for GDM as clinically indicated. • Refers to specialist clinic via e-Referral as needed and documents plan in the HHR and DMR. • Provide guidance to inpatient and outpatient staff on • insulin adjustments and ongoing glycaemic management. • Review caseloads and peer review as required.

Obstetric Consultants	• Provide the MAPS Midwife care plans for complex • GDM cases. • Determine the mode and timing of birth for women with GDM • on insulin or oral hypoglycaemic agents, or those with • suboptimal glycaemic control, fetal complications, or other • obstetric indications. • Review women as needed and document plan in the HHR • and DMR. • Provide care for women living with GDM who • require obstetric led care due to other • comorbidities / complexities. • Review caseloads and peer review as required.
General Practitioner Obstetrics (GPO)	• Supports coordinated approach with MDT members for • ongoing support of women diagnosed with GDM. • Refer to specialist clinic via e-Referral as needed and documents plan in the HHR and DMR. • Delivering safe and high-quality outpatient services. • Supports coordination of care as part of MDT. • Review caseloads and peer review as required.
Myrtle Diabetes Educators Midwives & Diabetes Service Educators	• Complete GDM education within 1 week of referral. • Register the woman with the NDSS (with consent) and document in the DMR. • Educate on blood glucose level (BGL) monitoring: four times daily (fasting and 1–2 hours postprandial) and target ranges. • Educate on the importance of exercise, with tailored plans to meet individual needs. • Educate on complications, insulin use, test results, • and potential maternal, fetal, and neonatal complications of GDM. • Refers to specialist clinic via e-Referral as needed and documents plan in the HHR and DMR. • Review BGLs and refer as needed. • Interpret diabetes test results and understand glycaemic targets. • Review caseloads and peer review as required.
Myrtle DE – Ongoing Care	• Weekly BGL review. • Refer to Endocrinologist for pharmacotherapy/insulin initiation if BGL targets are not met.

	• Offer referral to Lactation consultant for antenatal education on breastfeeding, including the benefits for women with diabetes • Refers to specialist clinic via e-Referral as needed and documents plan in the HHR and DMR. • Identify maternal complications and refer to Obstetric team. • Refer to Obstetric ANC if the woman is not adhering to management recommendations, Glycaemic control remains suboptimal, additional obstetric risk factors are present. • Ensure the woman is aware her neonate requires BGL monitoring for the first 24 hours post-birth. • Review caseloads and peer review as required.
Midwives (ANC, AAU, LBS MAPS, Postnatal, VMS)	• Provide antenatal education and care. • Conduct history taking and physical assessment as part of routine antenatal care. • Continue antenatal care for women with GDM controlled through diet and exercise, who do not present additional obstetric risks—these women may remain in the midwifery model of care. • Support and educate on the benefits of Breastfeeding for women with diabetes. • Encourage antenatal breast stimulation and expressing from 36 weeks' gestation, unless contraindicated. (Refer to Lactation clinic for outpatient appointment). • Identify complications and refer to appropriate clinician. • Ensure the woman is aware her neonate requires BGL monitoring for the first 24 hours post-birth. • Encourage postpartum screening for type 2 diabetes at 8-12 weeks postpartum.
Allied Health – Dietitians	• Educate on dietary management for GDM. • Guide healthy pregnancy weight gain. • Support balanced nutrition and long-term diabetes risk reduction. • Promote sustainable eating habits postpartum. • Provide dietary guidance to other professionals. • Participate in weekly MDT meetings.
Allied Health – Physiotherapy	• Educate on safe physical activity during pregnancy. • Tailor exercise programs for glycaemic control and maternal health. • Manage musculoskeletal conditions affecting mobility.
Allied Health – Psychological Services	• Provide psychological support for GDM-related distress. • Develop coping strategies for home management.

	- Assess and support mental health concerns (e.g. anxiety, depression). - Refer to perinatal mental health services as needed.
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12. GDM Education

All women must be provided with culturally appropriate education with materials tailored to the individual's health literacy level and access to interpreters or culturally aligned services as needed.

12.1 GDM Education

All women will receive education in relation to:

- What is diabetes? The types of diabetes
- Basic physiology - the role of food and insulin.
- The effects of diabetes on pregnancy.
- The effects of pregnancy on diabetes.
- The role of healthy eating in blood glucose control.
- The role of exercise in blood glucose control.
- Lifestyle changes.
- Postnatal issues.
- Advice regarding planning future pregnancy and contraception.
- Managing the psychological/ psychosocial aspects of coping with diabetes.
- Providing women access to equipment and supplies to effectively manage their diabetes.
- Self-monitoring blood glucose.
- Breastfeeding and GDM
- Future risk of diabetes
- Self-insulin administration and insulin adjustment (if required)

12.2 Insulin Education

All insulin education will include:

- Safe use of insulin
- Injection technique
- Management of hypoglycaemia
- Insulin pharmacology and effects

Table 7 Education for safe self-administration of Insulin

Aspect	Consideration
Safety	- Ideally provided by a diabetes educator or midwife trained in teaching self-administration of insulin - Can be individual or group sessions, noting that group sessions have been reported as positive due to peer support through a shared experience²³ - Confirm type of insulin and dose ordered
Demonstrate	- Insulin delivery device - Needle size

	• Applying needle to device • Priming device and dialling dose • Injection sites and rotation • Self-injection technique
Discussion points	• Hand washing • Insulin action and profile • Timing of injection • Recognition of hypoglycaemia symptoms and treatment • Refer to Table 6. Hypoglycaemia in women receiving glucose lowering medication • Potential side effects • Safe disposal of sharps • Safe driving ²⁴ • Storage and handling of insulin • Expiry of insulin (opened and unopened) • Confirm and update NDSS registration to enable access to free insulin needles • Travelling—NDSS card or letter authorising woman to carry insulin and needles in hand luggage ²⁵

12.3 Breastfeeding and Antenatal Expressing Education

There are specific benefits to breastfeeding for women diagnosed with GDM including but not limited to: stabilising neonatal blood sugars, reducing the neonate's future risk of obesity and diabetes, and significantly lowering the mother's risk of progressing to Type 2 Diabetes or metabolic syndrome.

Antenatal colostrum expression for women with GDM may be advised as a strategy to reduce newborn hypoglycaemia and formula use with careful instruction and monitoring. Women with GDM should be offered to commence antenatal expressing from 36 weeks' gestation (36+0), following standard practice and eligibility.^{8,9,10}

Breastfeeding education and support should be provided from 30 weeks' gestation by AKG Lactation Consultants via:

- Online group sessions
- Individualised appointments, as required

12.4 Physical Activity Education

Physical activity is a safe and effective adjunct therapy for GDM. Both aerobic and resistance exercises improve blood glucose levels and may reduce the risk of fetal macrosomia, caesarean delivery, and pharmacological intervention.^{16,17,18,19,20}

All women will be provided with education on suitable physical activity to do during pregnancy, any women requiring a more tailored approach will be referred to AKG Physiotherapy by the MAPS Myrtle GDM Midwife/DE.

See [Appendix 7](#) for a full overview of physical activity in GDM management.

13. GDM BGL review

All healthcare professionals involved in antenatal care share the responsibility of reviewing BGLs for women diagnosed with GDM. At each antenatal visit, the attending healthcare professional must review the woman's BGL records.

If BGLs are outside the target range, the attending healthcare professional should contact the AKG Dietitian or AKG Myrtle Diabetes Educator/Midwife for review during the antenatal appointment, where possible.

If an in-appointment review is not possible, the attending clinician must ensure follow-up by contacting the AKG Dietitian or AKG Myrtle DE via email or phone to arrange a review as soon as possible.

Women diagnosed with GDM will be allocated a series of scheduled appointments for submitting their BGLs via email.

Initial Review:

AKG Dietitians and DE's will conduct the initial BGL review following the completion of education.

Ongoing Monitoring:

Women will be scheduled for regular appointments and instructed to submit their BGLs via email for review.

Antenatal Clinic (ANC) Visits:

If a woman attends ANC, the practitioner conducting the visit is responsible for reviewing her BGLs.

Out-of-Target BGLs:

If BGLs are outside the target range:

- The AKG midwife, doctor, or midwifery coordinator must liaise with the AKG Dietitian or Myrtle Midwife/DE for a review during the ANC appointment, where possible.
- If a review cannot be conducted during the ANC appointment, the practitioner must promptly email or contact the AKG Dietitian or Myrtle DE to ensure the BGLs are reviewed as soon as possible.

13.1 Special considerations

- Ramadan - pregnant women are exempt from fasting during Ramadan if it poses a risk to their health; however, some may still choose to fast. Fasting during pregnancy has been associated with lower mean glucose levels and an increased risk of hypoglycaemia compared to non-fasting women⁶. If a woman chooses to fast, involve the healthcare team to develop an individualised care plan⁷.
- Shift workers – Adapt monitoring and medication to shift schedule. Encourage predictable mealtimes to maintain stable glucose levels.

13.2 Targeted range for BGL lifestyle treatment

- Fasting < 5.3
- 1 hour postprandial < 7.4
- 2 hours postprandial < 6.7

13.3 Women continuing to have BGL outside of target range

Women that continue to have BGL' outside of the target range must have the following implemented:

- Organise review by the AKG Dietitians
- Contact Myrtle DE – who will liaise with the Specialist Obstetrician or Endocrinologist to discuss treatment options and triage.
- If the woman has random BGL's out of target, considerations should be towards diet and exercise.
- Fasting – consistently three or more BGL's above 5.3 mmol/L. Once pharmacological treatment has commenced, adjustments will be based on the criteria of fasting < 5.1 mmol/L.
- Postprandial – consistently three or more above 6.6mmol/mL at 2-hour PP for the same meal i.e. X 3 breakfast BGL's in a row. If woman is testing BGL's at 1 hour three or more above 7.3mmol/L PP for the same meal. (This is not usual practice but necessary at times)

13.4 Targeted range for BGL Pharmacological treatment

- Fasting < 5.1
- 1 hour postprandial < 7.4
- 2 hours postprandial < 6.7

14. Care of the woman with GDM

AKG offers women diagnosed with GDM access to the MAPS model of care, supporting a continuity of midwifery carer model throughout pregnancy and the postnatal period.

14.1 Eligibility and Referral Pathways

- Women may be allocated to MAPS early in pregnancy based on:
 - Pre-existing risk factors
 - Previous GDM diagnosis
- Self-referral to MAPS is available from 8 weeks gestation.
- General Practitioners may also refer women to MAPS.
- Women diagnosed with GDM later in pregnancy (after initial care allocation) may:
 - Transfer into MAPS, or
 - Continue with their current antenatal care provider.

Regardless of the model of antenatal care chosen or allocated, all women receive equitable education, support, and clinical expertise tailored to GDM management.

14.2 Care Delivery Model

- Antenatal and diabetes care for women with GDM should be provided in multidisciplinary clinics, where possible.
 - Integrated model of care reduces the need for multiple appointments.
 - Enhances patient attendance, engagement, and coordinated management.
- Individualised appointments may be required to accommodate:
 - Language needs
 - Cultural considerations
 - Specific health or social needs

14.3 ANC Appointment Schedule

ANC appointments should follow the national Patient Handheld Record (PHHR) schedule, with consideration for additional visits as clinically required/indicated.

14.4 Obstetric review:

- Obstetric team will have face to face appointments with the woman at key points:
 - First obstetric appointment: to establish Obstetric care plan
 - 32–34 weeks: post-growth and wellbeing ultrasound for discussion of mode of birth. Can be earlier dependant on documented diabetic team management plans and timing of scans.
 - Early diagnosis GDM: 28-week overall.
 - Additional appointments as clinically required
- The treating team is responsible for documenting the management plan on the Obstetric Risk form and writing in the woman's Digital Medical Record (DMR).
- All post-dates' women, regardless of GDM status, require additional care in AAU. Refer to [AKG Antenatal Clinical Guideline](#) for post-dates management.

Table 8 Mode and timing of birth

Aspect	Consideration
Context	• There is limited strong evidence that informs timing and management of birth relating to expectant management, offering IOL or CS.^{28,29}
Timing of birth	• GDM women who have maintained target BGL's with no fetal macrosomia or other complications- wait for spontaneous labour unless other indications for IOL offer IOL at 41+3 weeks. • GDM – pharmacological therapy alone should not be an indication for birth before term i.e. 39 weeks.
Induction of labour	• There is no clear evidence that women with GDM carrying a normally grown fetus require different indications for induction of labour (IOL) compared to women without GDM.²⁹ • When discussing expectant management vs. IOL, consider coexisting risks (e.g. pre-eclampsia, growth restriction, obesity) that may increase stillbirth risk. • For IOL before 39+0 weeks: • Balance the uncertainty of ultrasound-diagnosed macrosomia with potential short- and long-term neonatal outcomes.^{30,31}

Mode of birth	• Estimated fetal weight (EFW): • <4000 g: Vaginal birth generally appropriate. • 4000–4500 g: Assess individual factors (e.g. maternal stature, obstetric history, prior macrosomia ± shoulder dystocia, accuracy of EFW). • >4500 g: Consider elective caesarean section ^{31,33}; counsel on risks and benefits. • Note: Evidence is insufficient to confirm that caesarean section reduces the risk of birth trauma.³⁴
Communication	• A woman-centred approach should guide discussions on intrapartum management, including any pharmacological therapies. Women should be given the opportunity to ask questions and receive clear information to support informed decision-making.

15. Pharmacological Management of GDM

15.1 Initiation Considerations for Pharmacological Management of GDM:

Pharmacological treatment is considered based on:

- Degree/pattern of hyperglycaemia (fasting and/or postprandial)
- Gestational age
- Fetal growth parameters (e.g. macrosomia, LGA, SGA), including abdominal circumference (AC) and amniotic fluid index (AFI)
- Maternal BMI and weight
- Individual factors (e.g. fear of needles, social vulnerability, risk of misuse)

15.2 Insulin Therapy:

- First-line pharmacological treatment for GDM not controlled by lifestyle/diet
- Insulin type and dosing:
 - Fast-acting insulin with meals
 - Long-acting insulin nocte for fasting BGL control
- Education on insulin use is provided via AHS Diabetes Service / MAPS
 - Group education sessions
 - Individual sessions if required
 - Delivered by DE's and GDM-trained midwives

15.3 Oral Hypoglycaemic Agents (OHA)

Metformin:

- Discussion required prior to prescribing:
 - Category C drug in Australia
 - Placental transfer
- Considered when insulin is declined or inappropriate

Treatment is decided by an AKG Obstetrician Physician/Endocrinologist after DE's have identified that there is a requirement for treatment when lifestyle changes have not maintained BGL's in target range.

Table 9 Suggested Insulin type

Abnormality	Suggested Insulin Type	Action Profile
Elevated fasting glucose	Single nighttime injection of intermediate or long -acting insulins*	Onset 1-2.5 hours Peak. 4-12 hours Duration 16-24 hours
Postprandial Hyperglycaemia	Pre-meal rapid acting insulin*	Onset 10-20 mins Peak 0.5-1.5 hours Duration 5 hours
Fasting and Postprandial hyperglycaemia	Basal-bolus insulin regimen Pre-meal rapid acting and nighttime intermediate-acting insulin *	As above for specific insulin type

*Refer to an Australian pharmacopeia for complete drug information

16. Management of hypoglycaemia

Hypoglycaemia is uncommon in women with GDM, particularly those not receiving pharmacological therapy. In the absence of symptoms of hypoglycaemia, confirm the accuracy of results prior to initiating treatment. If not receiving pharmacotherapy, and the BGL is greater than or equal to 3.5 mmol/L and without symptoms is normal in pregnancy and does not require treatment.

If woman is using a sensor, advise to always check with a finger prick before treating a Hypoglycaemic event as sensor can produce false hypoglycaemic events.

Table 10 Hypoglycaemia in women receiving glucose lowering medication

Aspect	Consideration
Definitions	Mild hypoglycaemia: - BGL less than 4.0mmol/L and - May or may not experience symptoms of hypoglycaemia Severe hypoglycaemia: - BGL less than 3mmol/L and - May be associated with confusion and potential loss of consciousness - Medical assistance maybe required to manage i.e. follow Medical Emergency Team process.
Causes	- Excessive physical activity - Too much insulin - Missed, delayed or inadequate carbohydrate intake - Alcohol intake decreases BLG (note not recommended in pregnancy)
Symptoms	- Hunger - Light headiness/headache - Trembling/shaking/weakness

	• Sweating • Lack of concentration • Behaviour change • Dizziness • Tearfulness/crying • Numbness around the lips/fingers • Irritability • Blurred vision
Treatment	Assess level of consciousness • If reduced, oral management contraindicated • Seek medical help Consume one 15 g serve of fast acting carbohydrates (one of the following) • 5–7 glucose jellybeans or • Glass of soft drink (not sugar-free) 150 ml or • Half a glass of fruit juice 125 ml or o Lucozade® 100 mL or • 3 heaped teaspoons of sugar or honey dissolved in water If after 15 minutes symptoms persist or BGL remains less than 4.0 mmol/L repeat one serve of fast acting carbohydrates • Do not over-treat with fast acting carbohydrates as this may lead to rebound hyperglycaemia When BGL is 4.0 mmol/L or above eat longer lasting carbohydrate: • Eat a snack (e.g. sandwich or crackers, glass of milk) or • Usual meal if within 30 minutes Avoid over treatment of hypoglycaemia resulting in hyperglycaemia • Document BGL, time of hypoglycaemic episode and any preceding factors
If Bariatric surgery	• Seek expert advice for management of hypoglycaemia • First line treatment options may not be appropriate post bariatric surgery
Hypoglycemia Prevention	• Eat regular meals with adequate carbohydrate serving • Always carry a food snack (including while exercising) • Aim to take long or intermediate acting insulin at the same time each day • Identify causal factors of the hypoglycaemic episode and avoid/mitigate for the future • Always carry blood glucose meter so BGL can be checked if symptoms present

17. Blood Glucose Monitoring & Follow-Up

17.1 Targeted range for BGL Pharmacological treatment

- Fasting < 5.1
- 1 hour postprandial <7.4
- 2 hours postprandial < 6.7

Women on pharmacological treatment (GDMI) must:

- Send in weekly BGL readings to AHS.MAPS.Myrtle@health.wa.gov.au

- Attend virtual “appointments” for BGL review

Follow-up system includes:

- SMS reminders via WebPAS
- Emails for non-compliance
- Next of Kin (NOK) contact if repeated non-responsiveness

AHS ADS DE /Myrtle DE is responsible for insulin titration using the AKMR60.8.2 Initiation and Stabilisation of Insulin Therapy in the Ambulatory Setting Form.

18. Psychological Considerations in Gestational Diabetes (GDM)

Pregnancy is a period of increased psychological vulnerability, and a diagnosis of GDM can further heighten emotional stress.¹¹ Women commonly report feelings of guilt, fear, isolation, judgment, and shame following a GDM diagnosis ¹¹.

Supporting emotional well-being is essential for optimal engagement in care and effective self-management.¹²

See [Appendix 5](#) for a full overview of psychological considerations in GDM

19. Care of the woman with GDM in the Antenatal Assessment Unit

All women will receive care in the Antenatal Assessment Unit (AAU) as outlined in the [AKG Antenatal Assessment Unit Guideline](#).

On arrival to AAU the primary midwife is responsible for

- Conducting a complete midwifery assessment and document findings on AKMR 8.
- Performing ongoing observations as clinically required.
- Conduct an admission BGL and blood ketones if admitted for BGL stabilisation.
- Undertaking routine BGL (Fasting and 2-Hour post prandial) and document on the AKMR 60.8.3 Obstetric Subcutaneous Insulin Order.
- Inform the on-call GPO of woman's presentation.
- Adhere to the appropriate AKG guideline/policy or procedure dependent on presentation.
- Refer to [Appendix 6](#) GDM/i patient presentation to AAU.
- Notifying the AHS Maternity Diabetic Educators of the presentation, management and plan. During business hours contact 9391 2901
- After-Hours the primary midwife must email the maternity diabetic educators to notify them of the presentation, management and plan on ahs.mapsmyrtle@health.wa.gov.au and ahs.adulthooddiabetesteam@health.wa.gov.au.

19.1 Targeted BGL for lifestyle

- Fasting < 5.3
- 1 hour postprandial < 7.4
- 2 hours postprandial < 6.7

19.2 Targeted BGL for Pharmacological treatment

- Fasting < 5.1
- 1 hour postprandial < 7.4
- 2 hours postprandial < 6.7

19.3 Escalation of Care in AAU – BGL $\geq 10.1\text{mmol/L}$ and/or Ketones are $\geq 0.6\text{mmol/L}$

Inform AAU Clinical Midwife and on call GPO. GPO to review patient and liaise with Specialist Obstetrician. Specialist Obstetrician / GPO create management plan in consultation with Myrtle DE. If required, after hours contact KEMH on-call Obstetric Physician via KEMH switchboard on 6458 2222 asking for Obstetric physician.

- BGL $>10\text{mmol/L}$ or/and Ketones are $>0.5\text{mmol/L}$ urgent RV by endocrinologist/Obstetric Physician
- GPO or Specialist Obstetrician should be involved in the formulation of the plan with the Endocrinologist/Obstetric Physician
- If Ketones are $>0.5\text{mmol/L}$ perform venous blood gas, identify if patient is in Diabetic Ketoacidosis and update GPO. Inform Obstetric Physician or Endocrinologist of results who will liaise with the specialist on call.
- Consider transfer to tertiary level care

19.4 Increased Surveillance and Timing of Birth for women with GDM

Table 11 Increased Surveillance and Timing of Birth for women with GDM

	GDM-Lifestyle	GDM-Pharmacological treatment	GDM-Lifestyle early diagnosis under 20 weeks gestation
Ultrasound Surveillance	• 32-34 week growth & wellness scan	• 32-34 weeks growth and wellness • Three (3) weeks post growth velocity If increased or decreased velocity re-scan minimum 3 weekly	• 28-week growth • 31-32 week growth velocity • If increased or decreased velocity re scan minimum 3 weeks
Timing of Birth	Offer IOL/CS 41+3/40 or as clinically indicated	Offer IOL/CS 39/40 or as clinically indicated	Offer IOL/CS 39/40 or as clinically indicated
Mode of Birth: • Consider offering IOL/CS if clinically indicated due to fetal size or additional obstetric complications • Woman to be at the centre of mode of birth discussion to ensure informed decision making. • Limited evidence to support offering or not offering IOL prior to 39/40 gestation			

19.4 Management of insulin and monitoring of BGLs for women undergoing induction of labour, elective caesarean section, and spontaneous onset of labour.

Table 12 Management of Insulin and monitoring of BGL's for women undergoing IOL, ECS and spontaneous onset of labour.

	BGL Monitoring	Insulin Regime
Induction of Labour GDM on Treatment	Continue with 4-point profile – fasting and two hours post - prandial until fasting.	- Continue long/intermediate acting insulin until established labour begins. - Continue short-acting insulin if the woman is eating; omit if not eating.
Elective Caesarean Section (ECS)	BGL on presentation to day surgery.	- Halve the dose of long/intermediate-acting insulin the night before surgery. - Practitioner Booking ELUCS to discuss insulin adjustments with woman at mode of birth discussion appointments and continue counselling by midwife or Doctor at all subsequent antenatal visits.
Spontaneous Labour	Continue with 4-point profile until fasting. Then 2 hourly	- Continue short-acting insulin if the woman is eating; omit if not. - Withhold all oral hypoglycaemic agents. - Consider the effects of long/intermediate-acting insulin given the night before.

20. Management of Women Requiring Betamethasone (Celestone)

Corticosteroids such as Betamethasone is used in women anticipated to have a pre-term labour and or birth, to reduce perinatal mortality and the incidence and severity of respiratory distress in neonates.

There are side effects from this therapy including hyperglycaemia due to the increase in insulin resistance. If hyperglycaemia occurs close to birth it will increase the risk of neonatal hypoglycaemia and hyperbilirubinemia. Diabetic Ketoacidosis may also be precipitated by corticosteroid administration in women with GDM.

Beta sympathomimetic agents such as Terbutaline should also be avoided if possible, during the period of steroid administration as they may exacerbate steroid induced hyperglycaemia.

20.1 Glycaemic management when receiving corticosteroids

See [Appendix 10](#) – Administration of Betamethasone during Pregnancy

- If possible, administer antenatal corticosteroids as early in the day as possible.

- Perform a baseline BGL prior to administration of the first dose of Betamethasone for all pregnant receiving Betamethasone:
 - Betamethasone Administration - No Risk Factors for GDM: Undertake BGL if < 6mmol/l provide written dietary advice
 - Betamethasone Administration – Risk Factors for GDM: Women who have not completed their OGTT during pregnancy but have risk factors for GDM, should have their BGL monitored following administration of Betamethasone and diet modification and or insulin therapy if required.
 - Betamethasone Administration – Women with diet-controlled GDM should continue to monitor their BGLs using 4-point profiles. During the first 72 hours after the initial dose of Betamethasone, carbohydrate intake should be limited to 2–3 serves per meal. If BGLs remain consecutively above target, insulin should be initiated.
 - Betamethasone Administration – GDM Requiring Insulin: These women may require their insulin doses increased by 50% over the course of the 72hours
- The primary midwife in AAU is responsible for contacting the MAPS DE or AHS ADS DE on 0434071993 during business hours to notify them of administration Betamethasone. The dietitian or DE is responsible for, providing education on carbohydrates and risk of BGL's increasing and the DE should discuss the plan of care with the MDT.
- If Betamethasone is started outside of hours and no plan documented on DMR – contact KEMH Obstetric Physician via switch board 6458 2222 for advice if required.
- Outside of business hours, the primary midwife must email the MAPS Myrtle Diabetic Educators on ahs.mapsmyrtle@health.wa.gov.au to notify them of the administration of Betamethasone so they can contact the woman to provide education and discuss BGL's and ensure the Endocrinologist/Obstetric Physician is aware to review woman.
- If the woman is being admitted as an inpatient, the midwifery team is responsible for commencing a 4-point blood glucose profile (fasting and 2 hours after all meals).
- The woman is admitted they will be under the GPO as per normal process with review by the Obstetric Endocrinologist/Physician and education and support from DE

21. Inpatient Antenatal Presentation

- During admission to hospital, all women with GDM should have an E-Referral to the AHS ADS completed. The Myrtle DE or AHS ADS DE will review the woman on Maud Bellas. During Business hours call the DE on and after hours email the DE's to notify them of the admission.
- Commence the woman on the AKMR60.8.3 Obstetric Subcutaneous Insulin Order.
- Women should be encouraged to use their personal BGL machine (if they have one) and equipment while an inpatient. If the patient requires further diabetes supply the midwife can contact the DE.
- Ensure the woman has adequate snack foods available in their room in case of a hypoglycaemic episode.

21.1 Inpatient Medication Orders

- On admission, prescribe the usual women's insulin regime on the AKMR60.8.3 Obstetric Subcutaneous Insulin Order form until the woman has been reviewed by the Obstetric Endocrinologist / Physician, or DE.
- Ensure OHA are charted on the patient medication chart.
- If there needs to be any adjustment to insulin as triggered by the subcutaneous insulin order form, then advice can be sought from Myrtle DE/ Endocrinologist during hours or KEMH Obstetric Physician out of hours if required.

- Contact the Myrtle DE/ Endocrinologist if you are in doubt regarding insulin orders. If out of hours contact KEMH Obstetric Physician via KEMH switch board 6458 2222.
- Insulin will be reviewed daily by the Obstetric Endocrinologist/ Physician, DE during weekdays, on weekends and public holidays. The management plan is to be documented in the DMR as a guidance. If on-call team identify that an adjustment needs to occur contact KEMH Obstetric Physician.
- Insulin and OHA should be reviewed -
 - When IOL or Caesarean Section is planned (See Birthing Section)
 - Woman is transferred to Birth Suite
 - BGLs are out of target range on more than 2 occasions at the same period.

21.2 Inpatient Antenatal Care (Midwifery)

- See [Appendix 9](#) – Admission for BGL stabilisation. Identifies standard observations required while inpatient. Please note that dependent on clinical situation observation schedule and fetal surveillance may need to be increased. Adhere to appropriate AKG policies/procedure and guidelines.

22. Care of the woman with GDMi in Birth Suite

22.1 Pharmacotherapy During Birth:

Management Plan

- The Specialist Obstetrician is responsible for consulting with the woman, between 34-36 weeks gestation to establish an individual management plan for labour and birth.
- All women who are GDM requiring insulin or OHA should ideally be booked first on the AM Elective Caesarean Section (ELUSCS) List.
- The treating team is responsible for documenting the management plan on the Obstetric Risk form and writing in the woman's Digital Medical Record (DMR).
- On arrival to Birth Suite, the Birth Suite Clinical Midwife and the Primary Birth Suite Midwife are responsible for reviewing the management plan, discussing the management plan with the GP Obstetrician and/or Specialist Obstetrician and implementing the management plan.
- At any stage during labour the Specialist Obstetrician may review the management plan and update the plan in consultation with the primary midwife, Birth Suite Clinical Midwife, on call GPO and woman.
- If blood glucose level of a woman who has been treated with insulin or OHA is 7mmol/l or over commence insulin infusion as outlined in AKMR 60.8.4. Notify the on-call GPO before contacting the Specialist Obstetrician and the AKG GDM Endocrinologist. If out of hours contact KEMH Obstetric Physician via KEMH switchboard 6458 2222 asking for the on call Obstetric Physician. The commencement of the infusion is to decrease the risk of the neonate developing hypoglycaemia in the immediate post-partum period.

22.2 Intrapartum Considerations

Induction of Labour (IOL)

- IOL in primiparous women may extend over a few days.
- Continue long/intermediate-acting insulin until established labour begins.
- Half the long-acting dose of insulin if dose is due before birth expected.
- Continue short-acting insulin if the woman is eating; omit if not eating.
- Monitor BGLs fasting and 2 hours PP until established labour, then every 2 hours.

Elective Caesarean Section (C-section)

- Halve the dose of long/intermediate-acting insulin the night before surgery.
- Practitioner Booking ELUCS to discuss insulin adjustments with woman at mode of birth discussion appointments and continue counselling by midwife or Doctor at all subsequent antenatal visits.
- Midwives and midwifery DEs are responsible for reinforcing this information.
- Day of admission BGL on admission and 2-hourly until birth.
- Offer food as soon as possible post birth.
- Monitor BGLs for 24 hours post-birth if treatment was insulin or OHA.

Emergency C-section

- Monitor maternal BGLs post-birth.
- Offer food as soon as clinically appropriate.
- Continue BGL monitoring via 4 PP for 24 hours if treatment was insulin or OHA.

Spontaneous Labour Onset

- Continue short-acting insulin if the woman is eating; omit if not.
- Withhold all oral hypoglycaemic agents.
- Consider the effects of long/intermediate-acting insulin given the night before.
- Offer food as soon as possible post-birth.
- Monitor BGLs via 4 PP for 24 hours post-birth if treatment was insulin or OHA.

22.3 Intrapartum BGL Monitoring – Women with GDM:

- Insulin requirements typically decrease during labour.
- Target BGL range: 4.0–7.0 mmol/L to reduce the risk of neonatal hypoglycaemia.
- Aim to maintain optimal glycaemic control while avoiding maternal hypoglycaemia.
- Assess clinical circumstances and reasons for high BGL (stage of labour, fasting status / recent juice/ food intake/ infection/ unwell).
- Consider repeating BGL in 1 hour / stat dose of subcutaneous insulin and reassess/ IV insulin infusion in consultation with Endocrinologist/ Obstetric Physician (BGL up to 8mmol/L may be acceptable if patient had just eaten/ other cause identified and expected to come down in next 1-2 hours).

During Induction of Labour

- Continue short-acting mealtime insulin (e.g. Novo Rapid) while eating, until fasting is advised.
- Check BGLs 4 hourly (fasting and 2 hours postprandial).

When Fasting Commences (labour or pre-op)

- Most women do not require insulin during labour or C-section.
- If BGL >8.0 mmol/L, consider IV glucose/insulin infusion (refer to infusion protocol).
 - Primary Midwife to escalate to Birth Suite Clinical Midwife.
 - Birth Suite Clinical Midwife to advise GPO and or Specialist Obstetrician, alongside DE / AKG GDM Endocrinologist to develop a management plan. Call KEMH on call Obstetric Physician if out of hours.
 - Commence insulin infusion as per AKMR 60.8.4.
 - Document on DMR the commencement of insulin infusion.
- If BGL <4.0 mmol/L, follow hypoglycaemia management protocol.
 - Primary Midwife to escalate to Birth Suite Clinical Midwife.

- Birth Suite Clinical Midwife to advise GPO and or Specialist Obstetrician, alongside DE / AKG GDM Endocrinologist to develop a management plan. Call KEMH on call Obstetric Physician if out of hours.
- Treat the hypoglycaemic event as outlined in MR 60.8.3 & Hypoglycaemia table 10
- Monitor BGL:
 - On arrival to hospital for C-section, then every 2 hours until birth and
 - Every 2 hours once established labour begins.

23. Insulin Infusion

Insulin infusion for GDM women is rarely required. If an insulin infusion is required, the MW should escalate their concerns to the GPO, who will review the patient and determine a management plan in liaison with the Specialist Obstetrician, Myrtle DE and AKG GDM Endocrinologist.

If out of hours contact the Obstetric Physician at KEMH via KEMH switchboard on 6458 2222 and request the on call obstetric physician.

Refer to AKMR60.8.4 Insulin Infusion Chart for Obstetric Women for protocols and infusion rates.

The below regime must only be applied for women in labour/caesarean section. It should not be used in the event of a hyperglycaemic crisis such as ketoacidosis, coma or hyperglycaemic syndrome.

Table 13 Insulin Infusion Preparation

Aspect	Consideration
IV Infusion	● Administer via infusion pump
Mainline	● Commence 1 litre 10% glucose at 50ml/hr
Sideline	● Require 50ml Syringe – Syringe Driver ● Add 50 units Actrapid (Short Acting) insulin in 50 ml of 0.9% Normal saline (i.e. 1 unit/ml) ● Prime infusion line – Before attaching to intravenous cannula, run the insulin/sodium chloride solution through the length of tubing twice to saturate the insulin binding sites on the plastic tubing.

Table 14 Initial Commencement Rate – Insulin Infusion

Blood Glucose Level	Rate of Insulin Infusion	Measure BGL
➤ 8 mmol/L	2ml/hr (2unit/hr)	1 Hour

24. Care of the woman with GDM postnatally

- Insulin is generally not required postpartum for women with GDM unless BGLs are consistently elevated.
- Ensure women are offered food soon after birth or on return to the postnatal unit
- Discontinue insulin immediately after delivery of the placenta if it was used during labour or caesarean section.

24.1 24-Hour BGL Monitoring for GDM requiring insulin or OHA Postnatally

- Women treated with insulin or OHA during pregnancy should complete a 24-hour BGL profile (fasting and 2 hours post each meal) prior to discharge. If woman wants to DC early 4PP can be completed at home with RV the next day by Myrtle DE or AKG Diabetes service.
 - If any BGL ≥ 10.0 mmol/L, notify the GPO and AKG Diabetes Educator.
 - Targeted Range For BGL: 4-8mmol/L.

24.2 Monitoring for GDM – Diet Controlled Postnatally

- BGL monitoring is not required in the postnatal period.

25. Care of the Neonate born to a GDM Diagnosed Woman.

Neonatal plasma glucose levels (PGLs) should be measured at the second and fourth feeds, both within the first 12 hours after birth. Infants should not go longer than 3 hours between feeds, and if expressed breast milk (EBM) is available, a post-feed top-up should be offered to reduce the risk of hypoglycaemia. Neonates must be closely monitored for signs and symptoms of hypoglycaemia during the first 48 hours; this can be undertaken at home, provided parents receive appropriate education and guidance. Confirm that the family is scheduled to receive a home visit on the day following discharge. For management of the neonate born to a GDM diagnosed woman refer to the following policies:

- [AKG Neonatal Management of Hypoglycaemia Guideline](#)
- [AKG Neonatal Jaundice Management Guideline](#)

Refer to [Child and Adolescent Health Service Neonatal Guidelines Hypoglycaemia](#) for ongoing management.

26. Long-Term Follow-Up – Women diagnosed with GDM

26.1 Postpartum Screening and Contraception

- Recommend a 6–8-week postpartum oral glucose tolerance test (OGTT). MAPS offer a 6–8-week post birth education session where women will be offered OGTT or HbA1c
- Repeat OGTT at least every 2 years thereafter.
 - The National Gestational Diabetes Register provides reminders to women and their GPs for follow-up diabetes screening.
- Discuss contraceptive options, considering risks, benefits, and individual preferences:
 - Progesterone-only methods are preferred for women planning to breastfeed.⁴⁵
 - Combined estrogen-containing methods may be introduced once breastfeeding is well established.
 - If other risk factors (e.g. hypertension) are present, consider IUD or progesterone-only agents.

26.2 Cardiometabolic Health

- Consider the increased long-term risk of metabolic syndrome.
- Reinforce the importance of lifelong screening for:
 - Cardiovascular disease.
 - Renal disease.

27. Future Pregnancy Planning

- Encourage women to plan future pregnancies in consultation with their healthcare provider.

- Offer support and advice on interpregnancy and preconception weight management:
- Each unit increase in BMI between pregnancies significantly raises the risk of recurrent GDM.
- Weight loss in previously overweight or obese women can reduce the risk of GDM recurrence by up to 80%.
- Recommend:
 - Preconception diabetes screening.
 - Preconception folic acid supplementation.
 - Early glucose testing in Referral and Communication.
- Provide timely written communication to the woman's primary care provider (e.g. GP), including:
 - Summary of maternal and neonatal outcomes.
 - Details of diabetes management during pregnancy.
 - Follow-up screening recommendations.
- Promote healthy lifestyle practices including:
 - A balanced diet.
 - Regular physical activity.
 - Weight maintenance or loss as appropriate.

28. Supporting information

The following documents support the implementation of this policy:

- HDWA
 - [Consent to Treatment Policy](#)
- EMHS
 - [Fetal Surveillance and Cardiotocography \(CTG\) Monitoring Policy](#)
- AKG
 - [Antenatal Assessment Unit Guideline](#)
 - [Adult Diabetic Ketoacidosis](#)
 - [Adult Diabetes Outpatient Procedure](#)
 - [Antenatal Clinical Guideline](#)
 - [Breastfeeding Policy](#)
 - [GP Obstetric Service Medical Protocol](#)
 - [Induction of Labour Guideline](#)
 - [Medication Management Procedure](#)
 - [Neonatal Feeding: Expressed Breast Milk – Identification, Storage and Thawing Guideline](#)
 - [Neonatal Management of Hypoglycaemia Guideline](#)
 - [Obesity in Pregnancy Guideline](#)
 - [Paediatrician Attendance for 'At Risk' Births Procedure](#)
 - [Risk Assessment - Identifying Moderate to High-Risk Women in Birth Suite Guideline](#)
 - [Small for Gestational Age and Fetal Growth Restriction Guideline](#)
 - [Ultrasound in Pregnancy Procedure](#)
- Other
 - [Child and Adolescent Health Service Neonatal Guidelines Hypoglycaemia](#)

29. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 1 - Clinical Governance Standard
- Standard 2 - Partnering with Consumers Standard
- Standard 4 - Medication Safety Standard
- Standard 5 - Comprehensive Care Standard
- Standard 6 - Communicating for Safety Standard

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31. Abbreviations

The following abbreviations are relevant to this policy:

Term	Abbreviation
AAU	Antenatal assessment unit
ADIPS	Australasian Diabetes in Pregnancy Society.
AHS	Armadale Health Service.
AKG	Armadale Kalamunda Group
ANC	Antenatal Clinic.
APT	Advanced Physician Trainee.
BGL	Blood Glucose Level.
BMI	Body Mass Index.
CALD	Culturally and linguistically diverse.
CEFM	Continuous Electronic Fetal Monitoring.
CNS	Clinical Nurse Specialist
CS	Caesarean Section.
DE	Diabetes Educator.
DM	Diabetes Mellitus.
DMR	Digital Medical Record.
EMHS	East Metropolitan Health Service.
ECS	Elective Caesarean Section.
FSH	Fiona Stanley Hospital.
GDM	Gestational Diabetes Mellitus.
GPO	General Practitioner, Obstetrics.
HHR	Pregnancy handheld record.
IGT	Impaired Glucose Tolerance.
IOL	Induction of labour.
KEMH	King Edward Memorial Health Service.
LARC	Long-Acting Reversible Contraceptive.
LBS	Labour and Birth Suite.
LGA	Large for gestational age.

Term	Abbreviation
LSCS	Lower Segment Caesarean Section.
MAPS	Midwifery Antenatal and Postnatal Service.
MDT	Multidisciplinary team.
MGP	Midwifery Group Practice.
MW	Midwife
NDSS	National Diabetes Services Scheme.
OGTT	75g 2-hour Oral Glucose Tolerance Test.
OHA	Oral Hypoglycaemic Agents.
PCOS	Polycystic Ovarian Syndrome.
QID	Quater-In-Die (four times per day).
RPH	Royal Perth Hospital.
VMS	Visiting Midwifery Service midwives who visit (mainly) postnatal women and newborns in the home environment.
WNHS	Women's and Newborn Health Service.

32. Document owner

Executive Director - Armadale Kalamunda Group

Enquiries relating to this guideline may be directed to:

Title: Director of Nursing and Midwifery
 Department: Executive
 Email: ahs.directorofnursingmidwifery@health.wa.gov.au

33. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V2	09 03 2026	09 03 2029	Early review, significant changes throughout document.

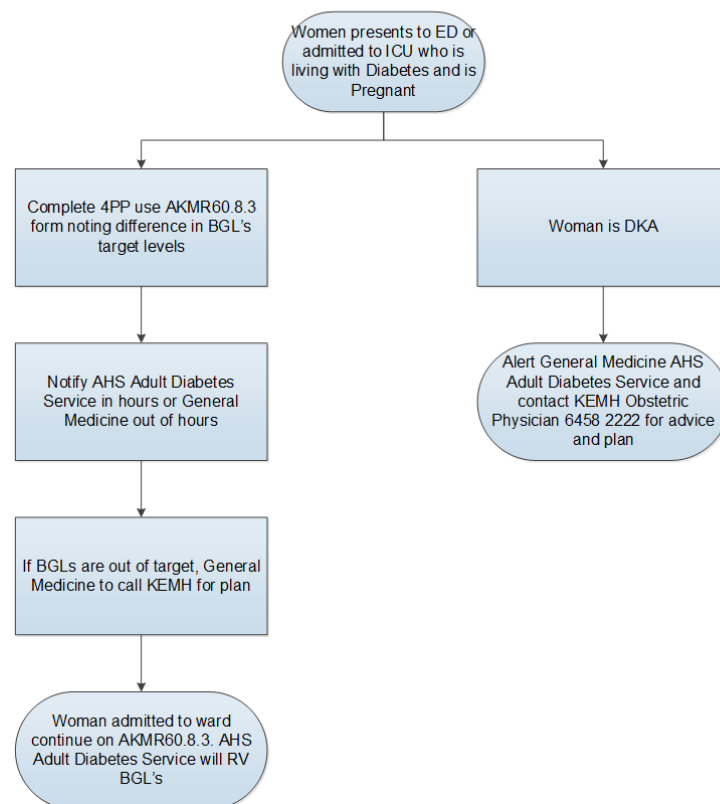
34. Authorisation

This guideline has been authorised and issued by the [enter position title of the authoriser].

Authorised by	Executive Director - Armadale Kalamunda Group
Authorisation date	05 06 2026
Published date	09 03 2026

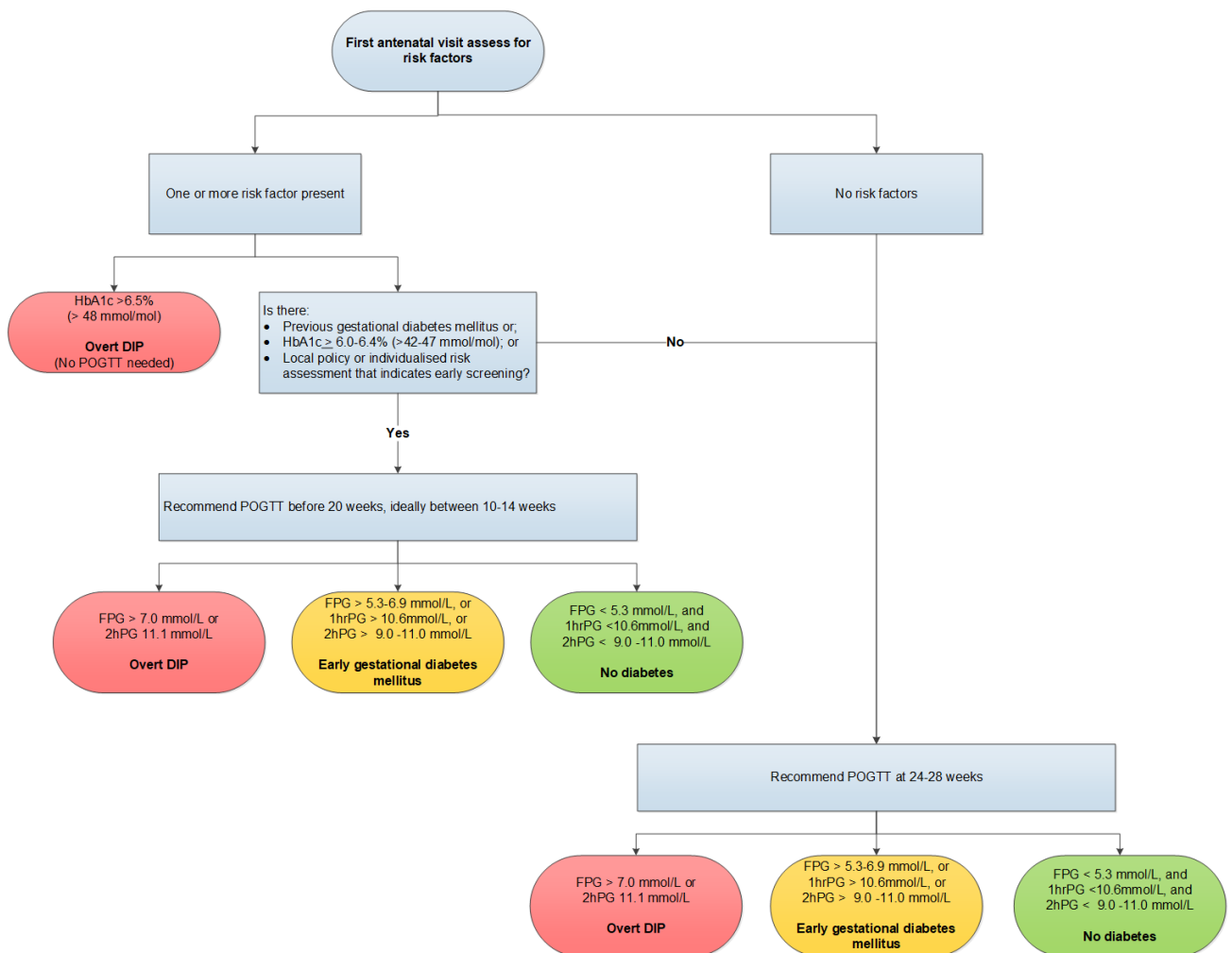
35. Appendix 1: Presentation and admission for non-obstetric condition of women living with Diabetes who is pregnant.

Presentation and admission for non-obstetric condition of women living with Diabetes who is pregnant.



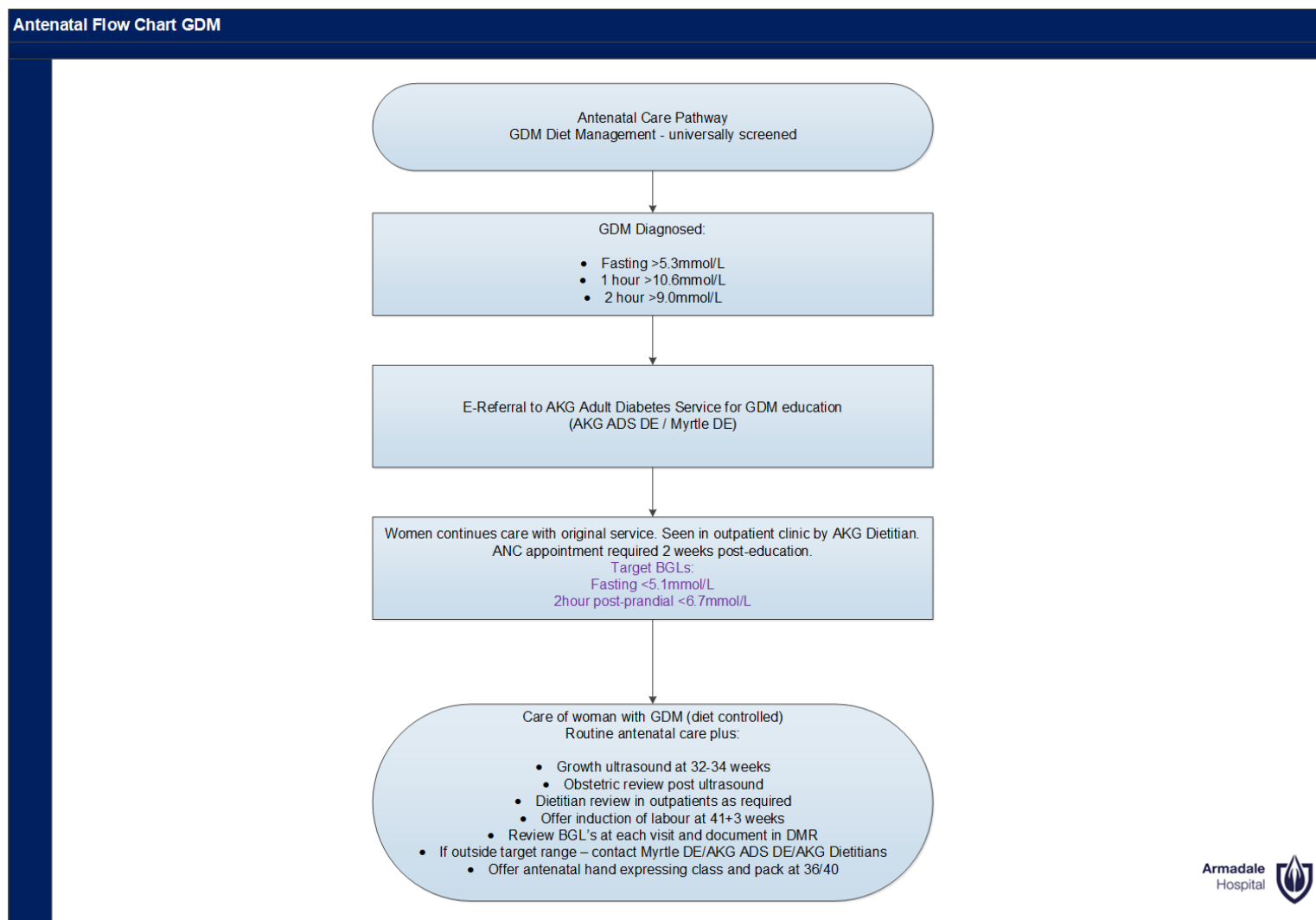
36. Appendix 2: Screening and Diagnosis of GDM

Screening and Diagnosis of GDM



ADIPS 2025

37. Appendix 3: Antenatal Flow Chart GDM



38. Appendix 4: Psychological Considerations in GDM

Pregnancy is a period of increased psychological vulnerability, and a diagnosis of GDM can further heighten emotional stress. Women commonly report feelings of guilt, fear, isolation, judgment, and shame following a GDM diagnosis.

Supporting emotional well-being is essential for optimal engagement in care and effective self-management.¹²

Barriers to treatment adherence may include:

- Depression and anxiety
- Disordered eating
- Stress
- Low health literacy
- Poor illness acceptance
- Conflicting information
- Financial constraints^{13,14,15}

Improving engagement may be supported by:

- Clear, empathetic communication
- Tailored education¹⁵
- Culturally appropriate resources
- Access to digital health technologies and tools¹⁵

Education considerations:

- Individualise care to align with the woman's values, preferences, and circumstances.¹¹
- Promote sustainable health behaviours that reduce long-term risk.
- Use evidence-based strategies such as:
 - Self-monitoring
 - Goal setting
 - Problem solving
 - Motivational interviewing

39. Appendix 5: Diabetic Ketoacidosis (DKA) in Pregnancy – Clinical Guideline (Abridged)

Aim: Support early identification and management of DKA in pregnancy. Enable staff to:

- Define hyperglycaemia in pregnancy.
- Distinguish DKA from hyperglycaemia and HHS.
- Respond to abnormal BGLs appropriately.

Immediate Actions:

- Notify the on-call Physician immediately WNHS/KEMH for any suspected DKA.
- Do not initiate extensive management prior to physician consultation.

Definition & Background

- DKA = Life-threatening complication of insulin deficiency.
- Triad: Hyperglycaemia, Ketonaemia, Metabolic acidosis.
- Common triggers: insulin omission, infection, hyperemesis, steroids, β 2-agonists.

Important: Pregnancy may mask classic DKA signs. Abdominal pain or contractions may be the only symptom.

Diagnosis Criteria

- History: Polyuria, polydipsia, vomiting, weight loss, abdominal pain.
- Labs:
 - BGL >13 mmol/L (can be lower in pregnancy)
 - pH <7.30 with $\uparrow$ anion gap
 - Positive serum or urine ketones
- Differential Diagnosis
 - Pancreatitis, lactic acidosis, salicylate toxicity, septic shock, HHS, appendicitis.
- Initial Assessment
 - Hourly BGL (capillary & lab)
 - ABG/VBG
 - Serum & urine ketones
 - Urea, electrolytes
 - ECG, cultures, CXR as indicated

Transfer & Communication

- Notify and involve: WNHS/KEMH Physicians, Medical Registrar, ICU, Duty Anaesthetist (if ICU admission).
- Transfer to ICU if pH <7.0, altered consciousness, or poor response.
- Treatment Overview
 - Fluid Resuscitation
 - 0.9% NaCl:
 - 1–2 L in first hour
 - 500–1000 mL/hr for 2–4 hrs
 - Total replacement over 24–36 hrs (50% in first 12 hrs)

Insulin Infusion

- Actrapid: 50U in 50mL 0.9% NaCl (1U/mL)
- Start at 6 mL/hr
- Continue long-acting insulin
- ABGs every 2–4 hrs
- Potassium Replacement
- <3.5 mmol/L: 10 mmol/hr IV
- 3.5–5.5 mmol/L: 20 mmol over 1–2 hrs
- >5.5 mmol/L: No replacement
- Monitor 1–2 hourly

Glucose

- Start 10% glucose at 40 mL/hr when BGL <10 mmol/L
- If BGL <7 or falling rapidly: increase glucose to 80 mL/hr

Other Considerations

- Catheter, NG tube, arterial line as needed
- Bicarbonate only if pH <7.0 and severe symptoms
- Phosphate only if <0.35 mmol/L with cardiorespiratory compromise
- Ceasing IV Therapy
- Stop insulin & fluids when:
- Ketones <0.5 mmol/L
- Acidosis resolved
- Patient eating normally

NB: Do not stop insulin if acidosis persists—even if normoglycemic. Increase glucose instead.

Transition to Subcutaneous Insulin

MDI patients: restart bolus with next meal + basal prior to ceasing IV.

Pump patients: Assess for pump failure; recommence after ketone clearance and oral intake.

Fetal Monitoring

- Start CTG monitoring if >28 weeks gestation.
- Continue until maternal stabilization.
- Fetal abnormalities (e.g., decelerations, minimal variability) usually resolve post-stabilization.
- Do not proceed with delivery solely due to FHR changes—stabilise mother first.

Consultation & Delivery Decisions

- Consult WNHS/KEMH Diabetes Service Obstetricians or on call Consultant for delivery decisions.
- Consider:
- Gestational age
- Fetal and maternal response to treatment
- Obstetric history
- Perinatal Risk
- Fetal mortality 10–36%

Risks: Preterm birth, fetal hypoxia, metabolic acidosis

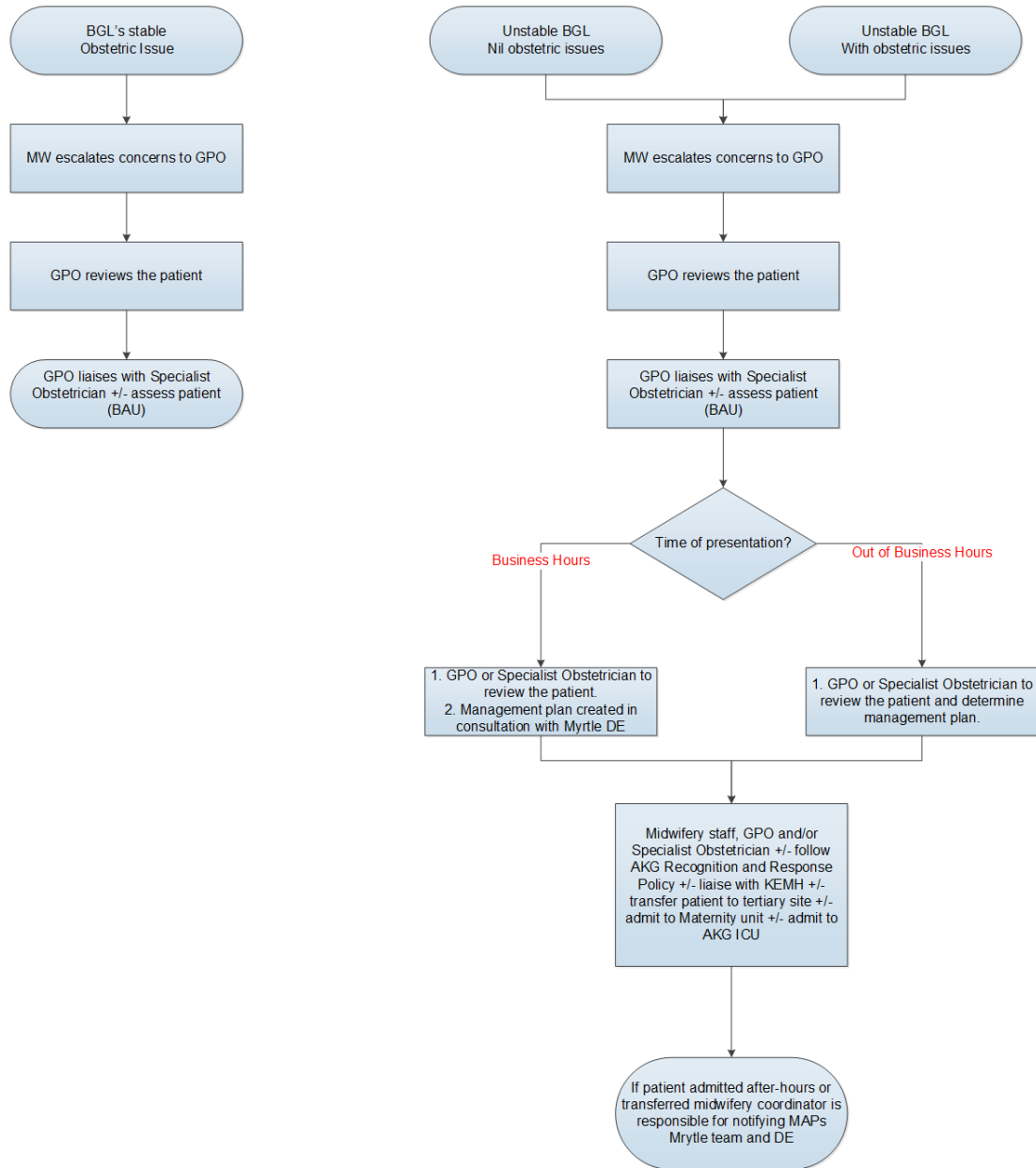
Table 15 DKA Protocol Summary

Infusion	Solution	Additive	Rate
Fluid Resus	0.9% NaCl	–	1–2L in 1st hr, then 500–1000mL/hr
Insulin	0.9% NaCl (50mL)	Actrapid 50U	6 mL/hr
Glucose	10% Glucose	–	40 mL/hr (↑ to 80 if needed)

Women who are pregnant and in DKA need to be discussed with the WNHS/KEMH team and potentially transferred to KEMH. Do not delay in contacting WNHS/KEMH treatment of DKA in pregnancy requires expert opinion.

40. Appendix 6: GDM/I Patient Presentation to AAU

GDMI Patient Presentation to AAU



41. Appendix 7: Physical Activity in GDM Management

Overview

Physical activity is a safe and effective adjunct therapy for GDM. Both aerobic and resistance exercises improve blood glucose levels and may reduce the risk of fetal macrosomia, caesarean delivery, and pharmacological intervention.^{16,17,18,19,20}

Recommendations

Type

- Aerobic: Walking, swimming, stationary cycling, prenatal classes²¹
- Resistance: Light/moderate (e.g. Pilates, bodyweight exercises)
- Modify activities as pregnancy progresses (especially in the 3rd trimester)

Duration

- Aim for 30 minutes on most, preferably all days of the week¹⁸
- Can be split into ≥10-minute moderate activity sessions²²

Intensity

- Tailor based on pre-pregnancy activity levels
 - Inactive: Start slow, gradually increase duration
 - Active: Maintain or reduce intensity
- Use RPE scale: Moderate intensity allows talking but not singing [Appendix 8](#)

Activities to Avoid

- Lying flat on the back (supine position)
- High risk of falling/trauma (e.g. contact or racquet sports)
- Exercises with bouncing/jumping (pelvic floor strain)
- Scuba diving or high-altitude activities

Contraindications

Do not recommend physical activity in women with:

- Significant cardiac or respiratory conditions
- Pre-eclampsia or placenta previa
- Short cervix, ruptured membranes, or preterm labour risk
- Vaginal bleeding or fetal growth restriction

When to Stop and Seek Medical Advice

Advise women to stop physical activity and seek care if experiencing:

- Chest pain, palpitations, dizziness, or severe breathlessness
- Headache, nausea, faintness
- Uterine contractions, vaginal bleeding, fluid leakage
- Reduced fetal movements
- Muscle or joint pain, swelling (calf, face, hands)

Additional Guidance

- Tailor activity to individual needs and clinical status
- Encourage hydration and avoiding exercise in hot/humid conditions
- Recommend daily activity tracking
- Refer to [Appendix 8](#) for exercise modifications and exertion guidelines

42. Appendix 8: Physical Exercise

Consider individual clinical circumstances when prescribing physical activity. Use the following heart rate ranges as a guide only.

Table 16 Target heart rate ranges for pregnant women

Maternal Age (years)	Fitness level or BMI	Heart rate range (beats/minute)
<20		140-155
20-29	Low	129-144
	Active	135-150
	Fit	145-160
	BMI > 25 kg/m ²	103-124
30-39	Low	128-144
	Active	130-145
	Fit	140-156
	BMI > 25 kg/m ²	101-120
40 +		125-140

Rating of perceived exertion (RPE) is a widely used and reliable indicator to monitor and guide exercise intensity. The scale allows individuals to subjectively rate their level of exertion during exercise or exercise testing.

Table 17 Ratings of perceived exertion

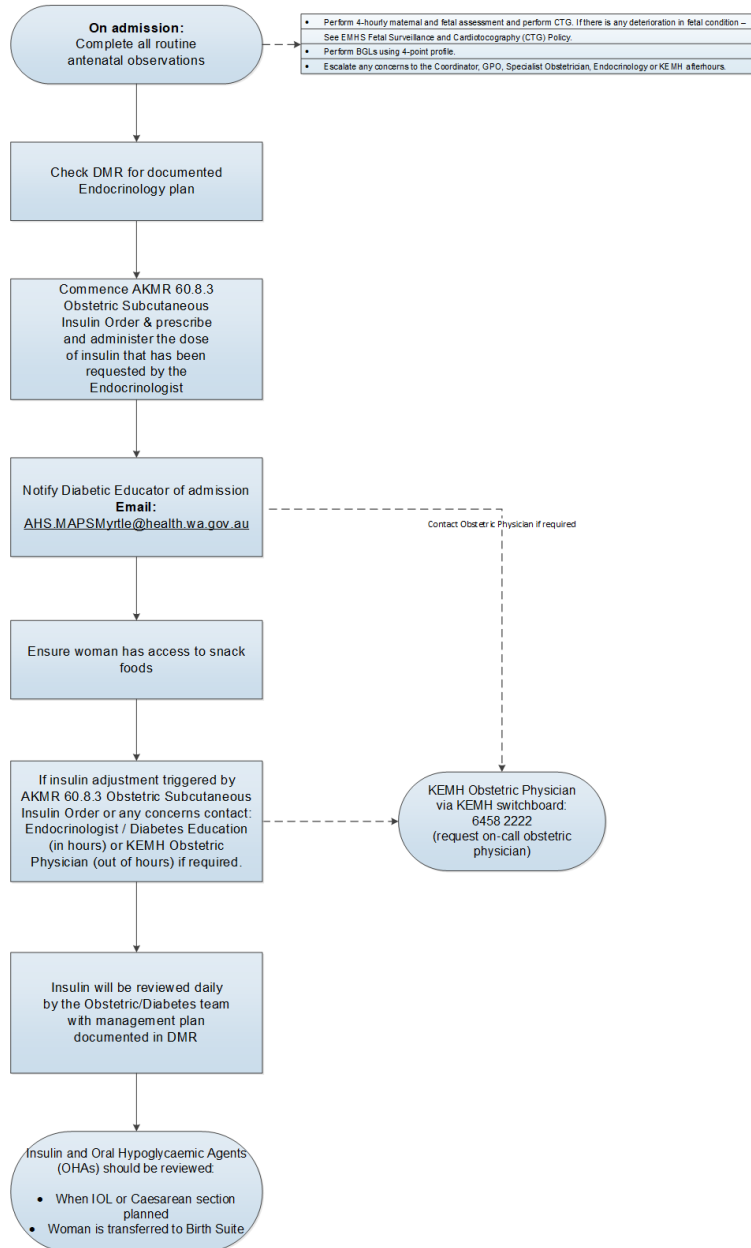
Rating of perceived exertion			Talk test
6		How you feel when lying in bed or sitting relaxed in a chair. Little or no effort	Can talk normally
7	Very, very light		
8			
9	Very light		
10			
11	Fairly light		
12	Somewhat hard	Target in pregnancy	Can talk but not sing
13		How you should feel with physical activity	
14			
15	Hard	How you felt with the hardest work ever done	Hard to talk
16			
17	Very hard		
18			
19	Very, very hard		
20	Maximum exertion	Don't work this hard	

Both charts adapted from: Sports Medicine Australia. Pregnancy and exercise. Fact sheet. n.d (cited 2024 December 18). Available from www.sma.org.au

43. Appendix 9: Admission – GDMi Women for BGL Stabilisation

Admission – GDMi women for BGL stabilisation

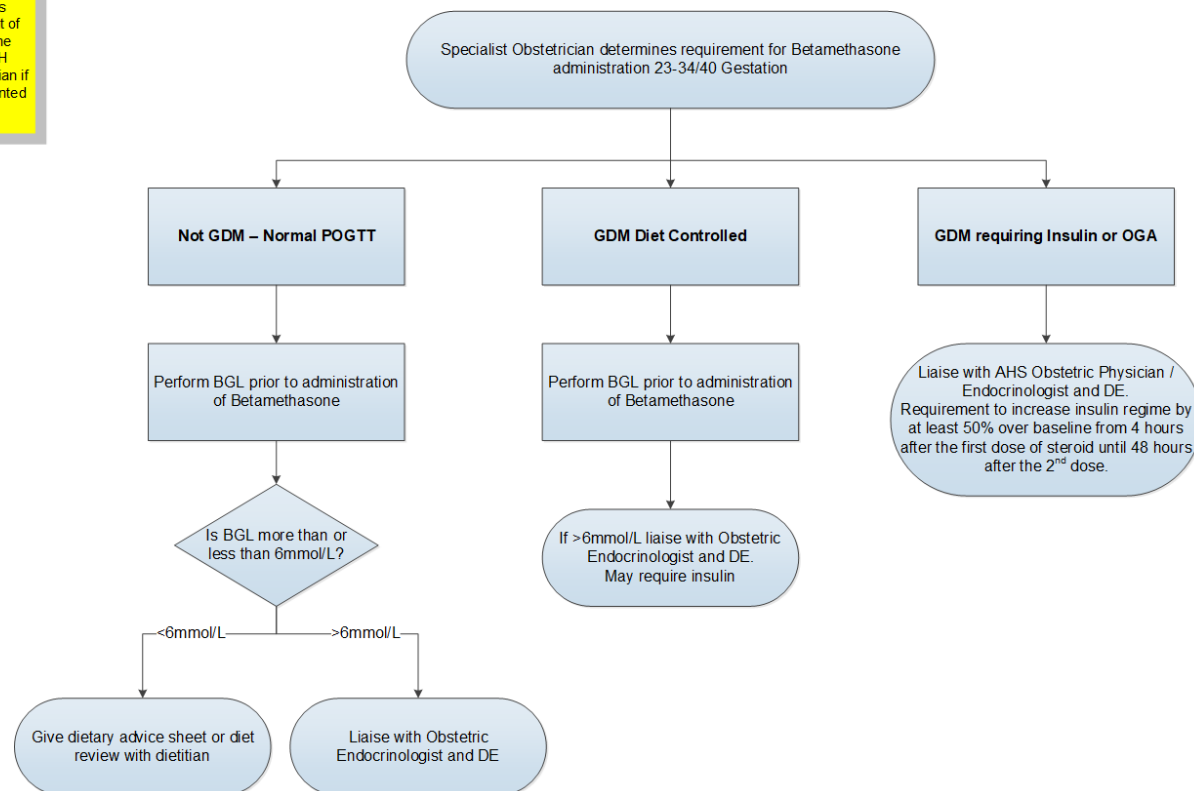
Note: These women would have been seen by the Endocrinologist prior to admission, i.e. planned admissions



44. Appendix 10: Administration of Betamethasone during pregnancy

Administration of Betamethasone during pregnancy

If out of hours commencement of Betamethasone contact KEMH Obstetric Physician if no plan documented in DMR



45. Appendix 11: GDM on insulin receiving Antenatal Corticosteroids total daily insulin dose < 40 units. Sliding scale insulin adjustment titrated against BGL readings.

Pre meal

BGL (mmol/L)	NovoRapid (units)
<5.0	0
5.0-7.0	4
7.1-9.0	5
>9.0	6 units and call Obstetric Endocrinologist/Physician

Post meal

BGL (mmol/L)	NovoRapid (units)
<7.0	0
7.0-9.1	4
9.0 -11.0	5
>11.0	6 units and call Obstetric Endocrinologist/Physician

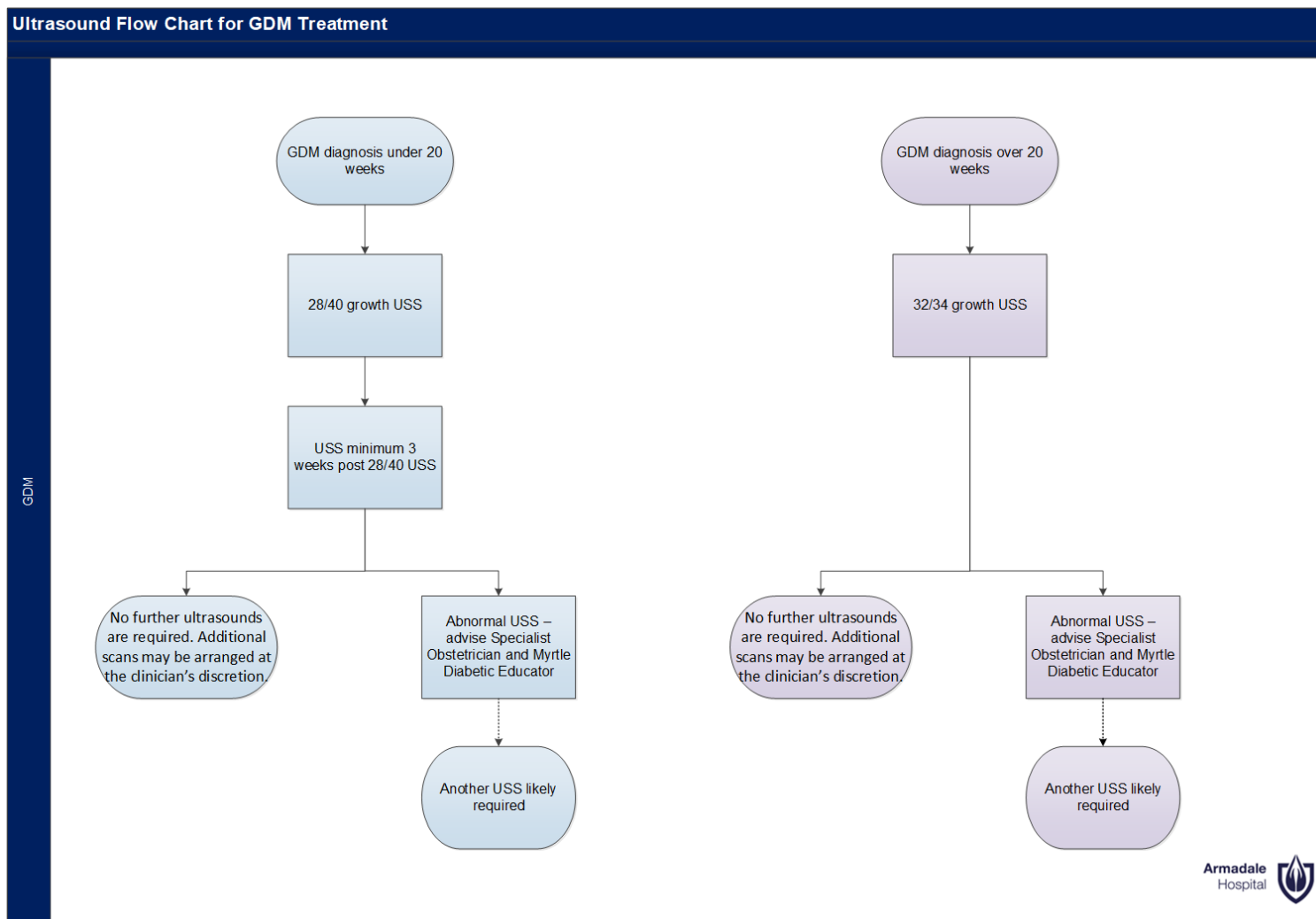
GDM on insulin receiving Antenatal Corticosteroids current total daily insulin dose < 40 units. Sliding scale insulin adjustment titrated against BGL readings.

BGL (mmol/L)	NovoRapid (units)
<5.0	0
5.0-7.0	6
7.1-9.0	8
>9.0	10 units and call Obstetric Endocrinologist/Physician

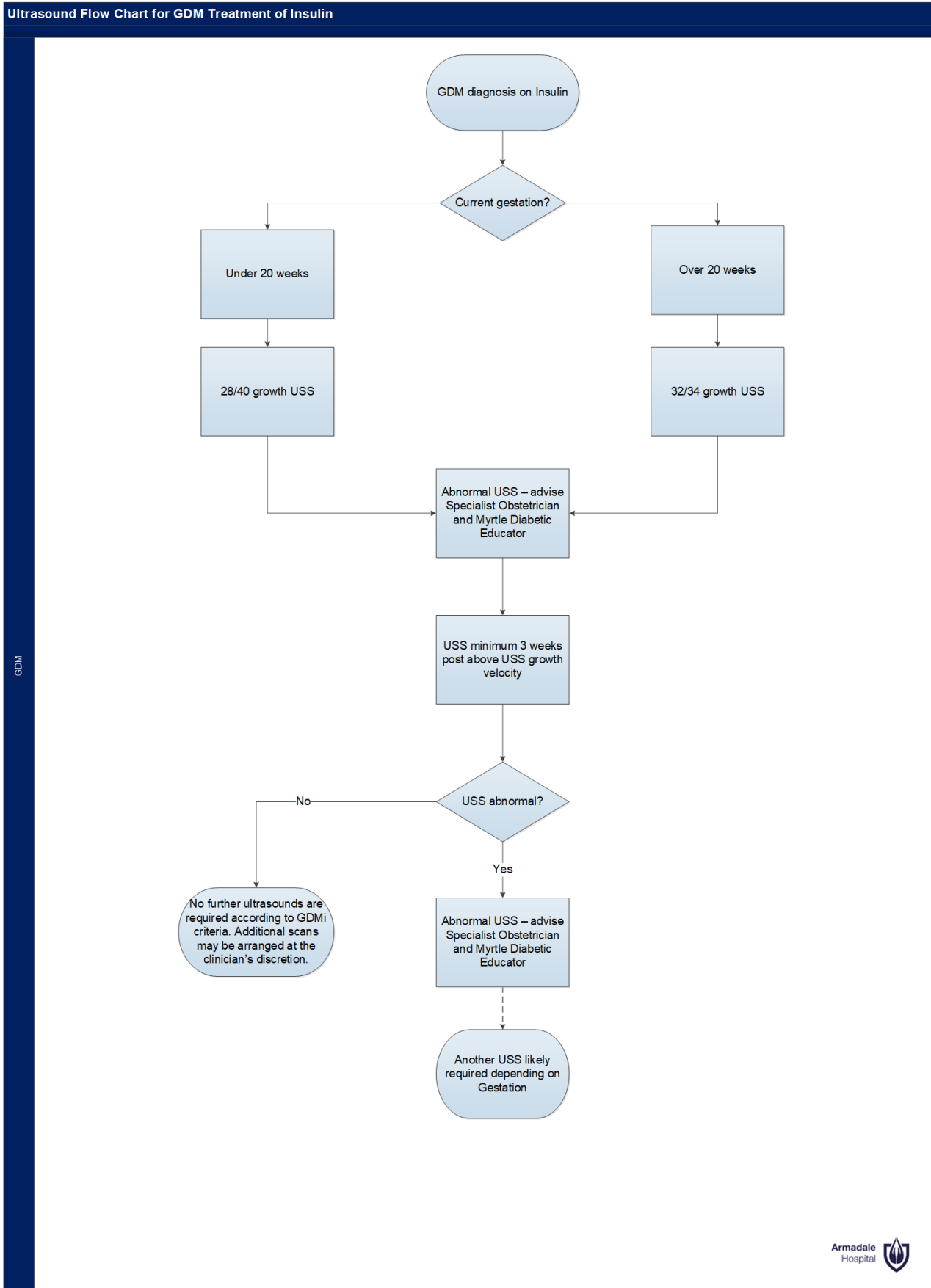
Post meal

BGL (mmol/L)	NovoRapid (units)
<7.0	0
7.0-9.1	6
9.0 -11.0	8
>11.0	10 units and call Obstetric Endocrinologist/Physician

Appendix 12: Ultrasound flow chart for GDM treatment

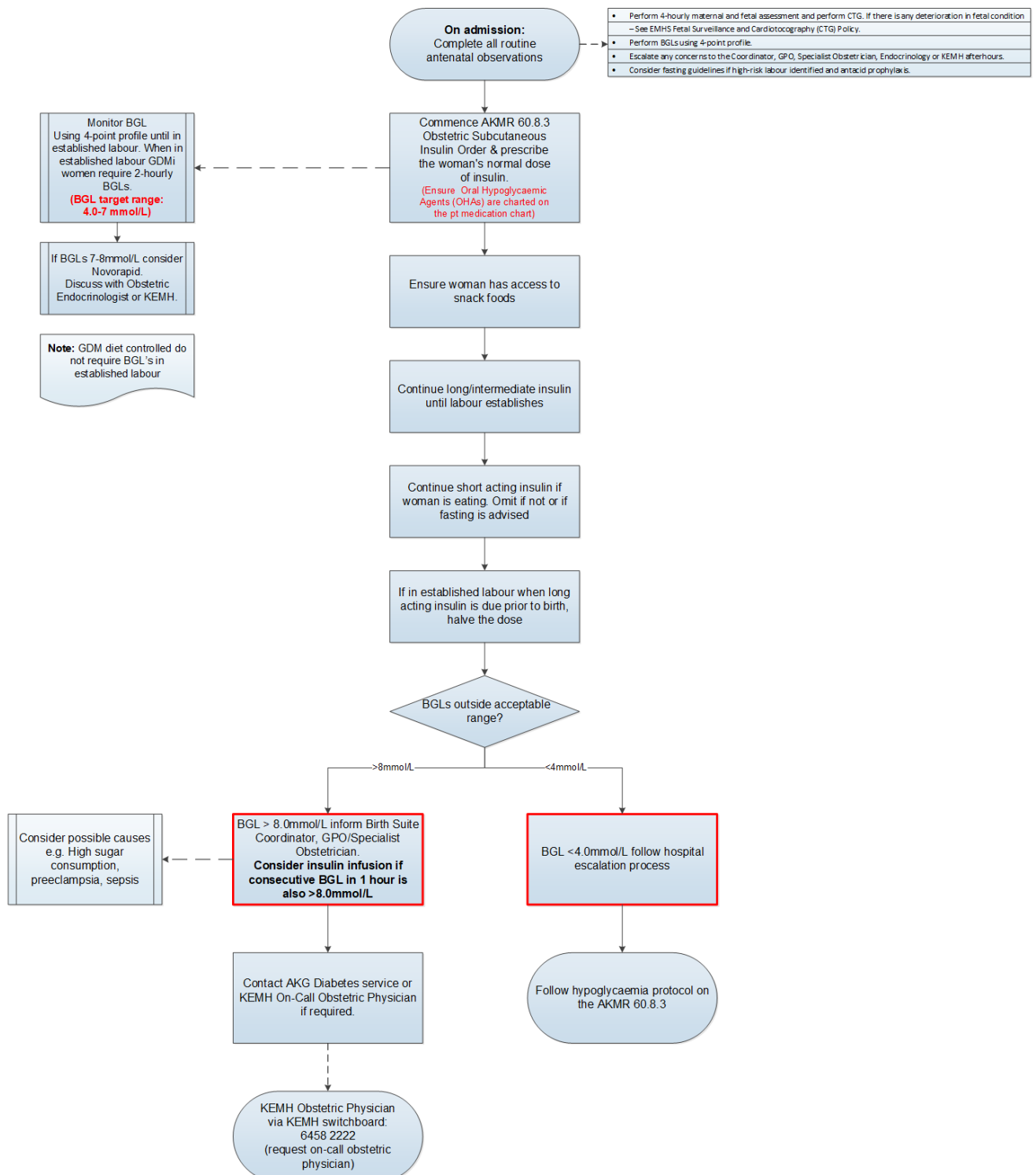


Appendix 13: Ultrasound flow chart for GDM treatment of insulin

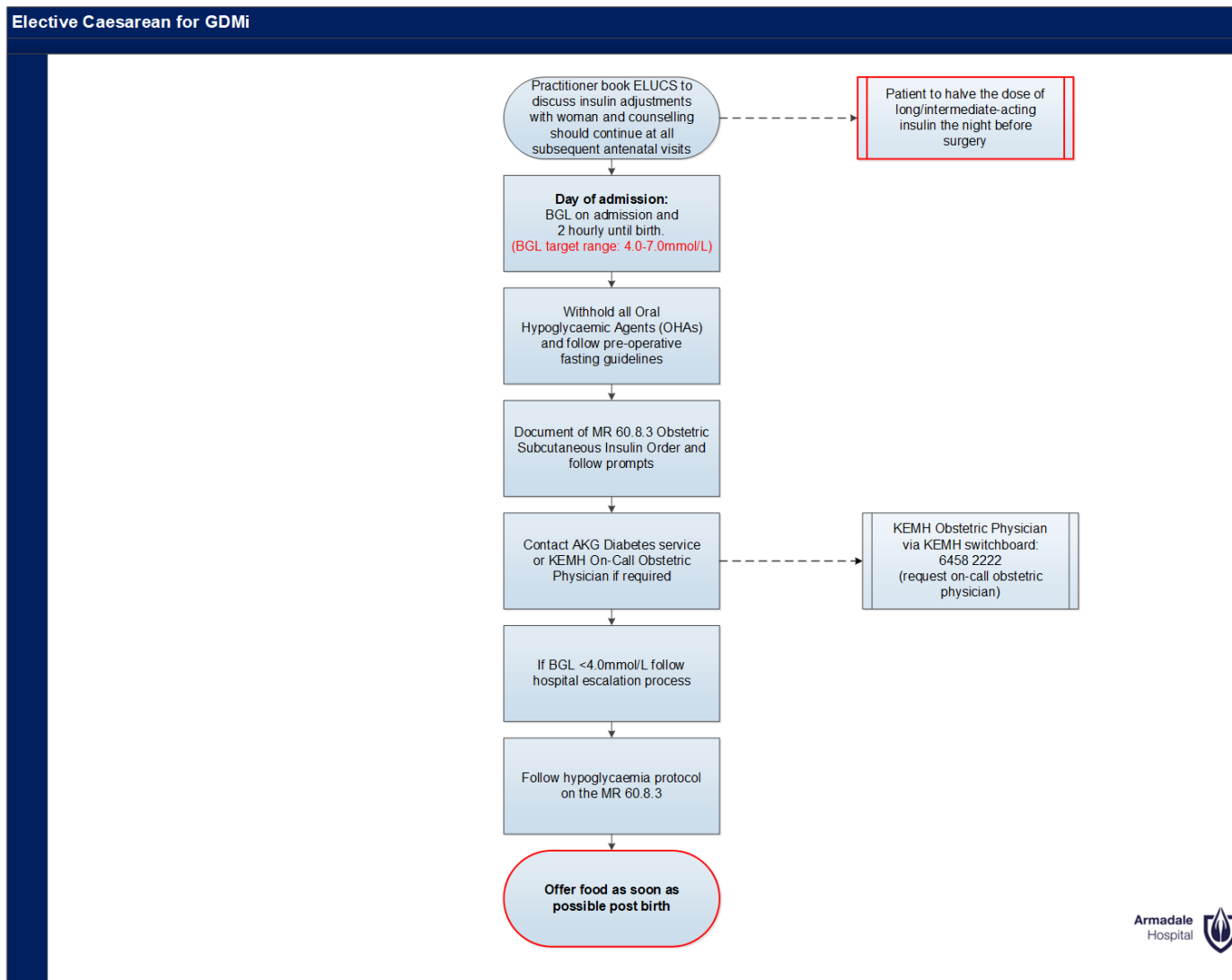


Appendix 14: GDM/GDMi management for induction of labour and spontaneous labour

GDM/GDMi Management for induction of labour and spontaneous labour

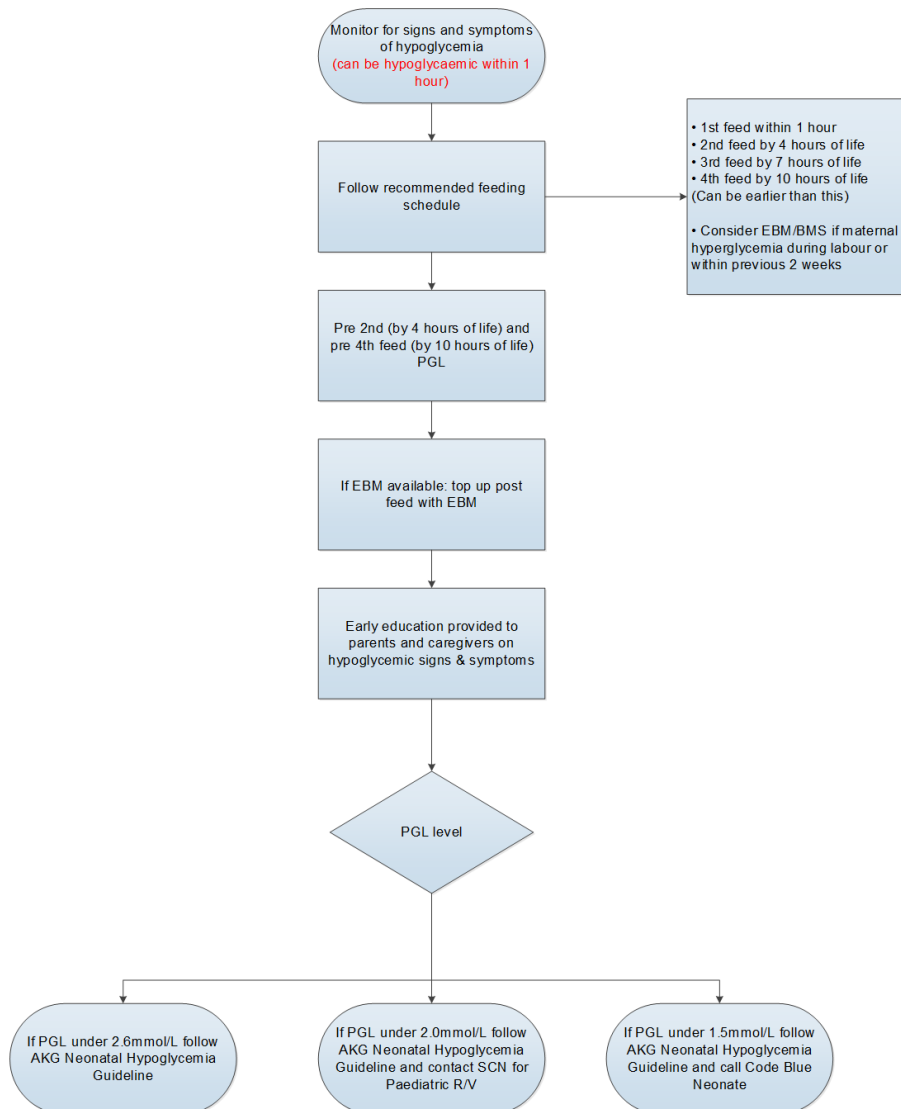


Appendix 15: Elective Caesarean for GDMi



Appendix 16: Care of the Neonate: GDMioRef

Care of the Neonate:GDM/i

**Note:**

• If maternal BGL's above 7 – 8 mmol/L may need treatment if continue to be elevated but not above 8mmol/L.

• It is important for the neonate that the maternal BGL's are within the lower target to avoid hyperglycemia.

• If not within target then there must be a neonatal plan of early feeding + EBM top ups etc.

• Neonatal insulin production takes 24-48 hrs to regulate depending on how hyperglycaemic the environment has been.

Appendix 17: Postnatal Care for GDMi women

